STATISTICS AND PROBABILITY – STA301 (QUIZ) Lectures 36-40

1. Sample proportion is an/are ___ estimator of population mean.

- None of these
- Unbiased — Lect. 33, page no. 255 Handouts
- Consistent
- Unbiased ad consistent estimator

2. Which of the following is most important and most widely used method in point estimation?

The most often used, **the maximum likelihood method**, uses differential calculus to determine the maximum of the probability function of a number of sample parameters. The moments method equates values of sample moments (functions describing the parameter) to population moments.

- The method of moments
- The method of fractional moments
- The method of maximum likelihood — Lect. 34, page no. 258 Handouts
- The method of least square

3. A statistic whose standard error decreases with an increase in the sample size will be:

- Unbiased
- None
- Consistent — Lect. 33, page no. 255 Handouts
- Efficient

4. We can apply method of Maximum Likelihood on:

- Discrete variables only
- Discrete as well as continuous variables — Lect. 34, page no. 258 handouts
- Continuous variables only
- Qualitative variables only

5. β is the probability of:

- Accept H0/H0 is true
- Accept H0/H0 is false — Lect. 36, page no. 273 handouts
- Reject H0/H0 is true
- Reject H0/H0 is false

6. Accept H0/H0 is false, is the _____.

- Correct decision
- Type I error
- None of the above
- Type II error — Lect. 36, page no. 273 handouts

7. Conventionally, the probability of making a type-II error is denoted by:

- Beta — Lect. 36, page no. 273 handouts
- Sigma
- Alpha
- Theta

8. _____ are equivalent:

- Type-II error and level of significance
- Type-II error and level of confidence
- Type-I error and level of significance — Lect. 36, page no. 273 handouts
- Type-I error and level of confidence

9. In general, the estimators obtained by the method of moments are _____.

- All of the above
- Inconsistent
- Biased
- Inefficient — Lect. 34, page no. 258 handouts

10. The larger the n, the smaller the standard error, and so the narrower the:

- None of the above
- Confidence level
- Level of significance
- Confidence interval — source Google

11. If we have the following information:
p=0.16, q=0.84, z=1.96 and e=0.05, then sample size will be:

- 207 — $n = z^2.p.q/ e^2$
- 200
- 205
- 199

12. “A point estimate plus/minus a few times the standard error of that estimate”. This statement represents:

- Confidence interval — Lect. 35, page no. 265 handouts
- Critical value
- Acceptance region
- Critical region

13. The maximum likelihood estimators (MLE) are ____ and ____ but not necessarily____.

- Unbiased, consistent, efficient
- Consistent, efficient, unbiased — Lect. 34, page no. 259 handouts
- Unbiased, efficient, consistent
- Consistent, unbiased, efficient

14. Inferential statistics involves

- Testing of hypothesis
- Estimation
- Confidence interval
- All of above — Lect. 36, page no. 272 handouts

15. Method of moments for estimation of parameters is based on the theory of

- Both linear and non linear
- **None of the above** **(Not sure)**
- Linear estimation
- Non- linear estimation

16. If Var (mean1) < Var (mean2) it implies

- Mean 1 is more consistent
- Mean 2 is more consistent
- Mean 2 is more efficient
- Mean 1 is more efficient Lect. 34, page no. 256 handouts

17. An estimator is said to be efficient if it has

- Smallest variance Lect. 34, page no. 256 handouts
- Unbiased estimate
- Both a and b
- None of these

18. In binomial distribution the sample size is considered to the sufficiently large, if both np and nq are greater than or equal to:

- 10
- 3
- 5 Lect. 30, page no. 230 handouts
- 15

19. Which of the following relationship exist between sampling distribution of samples mean and sampling distribution samples median?

- Var(mean)<var(median) Lect. 34, page no. 256 handouts
- Both of these
- None of these
- Var(median)<var(median)

20. Alpha is probability of _____

- Making type II error
- Rejection H1 when it is wrong
- Making type I error Lect. 36, page no. 273 handouts
- Accepting Ho when it is true

21. How can we interpret the 90% confidence interval for the mean of the normal population?

- There are 10%chances of falling true the value of the parameters
- All are correct
- There are 90% chances of falling true the value of the parameters (page no. 263 handouts)
- There are 100% chances of falling true the value of the parameters

22. An estimator is said to be _____ if its expected value is equal to true value of its parameter.

- Efficient
- Biased
- Unbiased — Lect. 33, page no. 253 Handouts
- Consistent

23. Sample proportion is a/an_____ estimator of population mean

- None of the above
- Unbiased — Lect. 34, page no. 255 handouts
- Consistent
- Unbiased and consistent estimator

24. A static whose standard error decreases with an increase in the sample size will be____.

- Unbiased
- None
- Consistent — Lect. 34, page no. 255 handouts
- Efficient

25. To determine a sample size which test we use _____.

- **Z** — **(Sure)**
- T
- None of these
- F

26. If Var (T1) > Var (T2), where T1 and T2 are two unbiased estimators then;

- T1 is more consistent
- T2 is more efficient — Lect. 34, page no. 256 handouts
- T2 is more consistent
- T1 is more efficient

27. The end points that bound the confidence interval are called:

- Upper limits
- Bounded limits
- Lower and upper limits — Lect. 35, page no. 262 handouts
- Lower limits

28. When sample size is to be considered large, population standard deviation can be replaced by:

- Sample standard deviation — Lect. 38, page no. 280 handouts
- Population variance
- Sample mean
- Sample variance

29. The confidence intervals become wide and less precise by:

- None of these
- Decreasing sample size — Lect. 35, page no. 265 handouts
- Decreasing the level of significance
- Increasing the sample size

30. In a Poisson distribution, Maximum Likelihood Estimator (MLE) for meu (μ) is equal to:

- Sample variance
- Reciprocal of the sample variance
- Sample mean — Lect. 34, page no. 259 handouts
- Reciprocal of the mean

31. In a Geometric distribution, Maximum Likelihood Estimator (MLE) for proportion (p) is equal to:

- Sample mean
- Sample variance
- Reciprocal of the mean — Lect. 34, page no. 259 handouts
- Reciprocal of sample variance

32. In a random sample of 500 men from Lahore city, 300 are found to be the smokers, the proportion of smokers is equal to:

- 0.4
- 0.3
- 0.6
- 0.5

33. Which of the following is not criterion for a good problem situation?

- It must be clearly structured and defined — (Confirmed from Riz Mughal File)
- It must be appropriate to the students' level of development (Not Sure)
- It must be authentic
- It must be such that solution will be benefited by group effort

34. The method of Maximum Likelihood (ML) is used to find out:

- Sample values
- Random numbers
- Correlation coefficient
- Point estimates — Lect. 34, page no. 256 handouts

35. If we draw a sample of size 'n' from a population the sample is called 'large sample' if:

- n>40
- n>30 — Lect. 32, page no. 248 handouts
- n>50
- n>25

36. Sample mean is a/an _____ estimate of population mean.

- Biased
- Unbiased — Lect. 34, page no. 254 handouts
- Either unbiased or biased
- None of the above

37. The width of the confidence interval is called ____ of the estimate.

- Unbiasedness
- Precision — Google, conceptual
- Accuracy
- Biased

38. The precision of an estimator can be increased by increasing the:

- Size of sample — (Google)
- Number of estimators
- Size of population
- Number of parameters

39. “A failed student is passed by an examiner”, is an example of:

- Type-I error
- Type-IV error
- Type II error — Lect. 36, page no. 273
- Type-III error

40. An experiment was conducted to estimate the mean yield of a new variety of wheat. A sample of 20 plots gave a mean yield of 2.9, and a 95% confidence interval of (2.48, 3.32). This means:

- We are 95% confident that the true mean yield of this variety is between 2.48 and 3.32

Lect. 35, page no. 265 (Concept taken from example-1)

- We are 95% confident that the true mean yield of this variety is 2.9
- None of these
- We are sure the true mean of this new variety is between 2.48 and 3.32

41. An alternative hypothesis is generally denoted by:

- H2
- H1 — Lect. 36, page no. 273
- H3
- H0

42. Ideally, the width of confidence interval should be:

- 0 — Lect. 35, page no. 265
- 1
- 100
- 98

43. A _____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

- Hypothesis
- Confidence interval — Lect. 34, page no. 259 handouts
- Probability limit
- Lower limit

44. Conventionally, the probability of making a type-I error is denoted by which of the following symbol?

- Theta
- Alpha — Lect. 36, page no. 273
- Sigma
- Beta

45. In interval estimation we obtained a ____ of values as an estimate of parameter.

- Group
- Range — Lect. 34, page no. 259
- Both a and b
- None of these

46. As sample size goes up, what tends to happen to 95% confidence interval?

- They become more narrow
- They become wider
- They become more precise and narrow — Conceptual & Google
- They become more precise

47. A failing student is passed by an examiner. It is an example of:

- Correct decision
- Type I error
- No information regarding student exams
- Type II error — Lect. 36, page no. 273

48. The precision of an estimator can be increased by decreasing the:

- Size of sample
- Number of estimators
- Number of parameters
- Level of significance — Conceptual, confirmed

49. If an estimator is more efficient then the other estimator, its shape of the sampling distribution will be

- Flattered
- Skewed of right
- Normal
- Highly peaked — Lect. 34, page no. 256

50. A confidence interval has a specified probability _____ of containing the true value of parameter.

- 1 – α — Lect. 35, page no. 263
- α
- 1 + α
- α /2

51. Interval estimation and Confidence interval are:

- Same
- Different — Lect. 34, page no. 259
- Dependent
- Independent

52. From which of the following methods we can obtain point estimate of the population parameters.

- Method of Moments
- All of the above — Lect. 34, page no. 256
- Method of Least Squares
- Maximum Likelihood Method

53. Which of the following statements about confidence intervals is inaccurate? A confidence interval for a mean always contain the sample mean

- If the population standard deviation increases, the confidence interval decreases in width

Google & Lect. 235, page 265 Handouts

- If we keep the sample size fixed, the confidence interval gets wider as we increase the confidence coefficient
- If we keep the confidence interval fixed, the confidence interval gets narrower as we increase the sample size

54. A deserving player was not selected in the cricket team. It is an example of:

- Type-II error
- Type-I error — (Lecture no 36 Page no 273 Handouts)
- Type-IV error
- Type-II error

55. An industrial designer wants to determine the average amount of time it takes an adult to assemble an "easy to assemble" toy. A sample of 16 times yielded an average time of 19.92 minutes, with a population standard deviation equal to 5.73. Assuming normality of assembly times, what is 95% (…)

the 95% C.I. for μ is given by

$$\overline{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

- (17.1123, 22.7277)
- (17.1123, 30.7277)
- (17.1123, 24.7277)
- (14.1123, 22.7277)

56. In testing H0: μ =100 against H1: μ ≤ 100 at the 10% level of significance, H0 is rejected if:

- The p-value is greater than 100
- The p-value is less than 100
- 100 is contained in 90% confidence interval
- The value of the test statistic is in the acceptance region

57. Confidence interval is also known as:

- Level of significance
- Margin of error
- Point estimator
- None of these Conceptual, Google

58. When an interval is constructed, what is the probability of occurrence of parameter in it?

- 50
- 1
- 0
- Either 1 or 0 (Page no 263)

59. The larger the standard error, the _____ the confidence interval.

- Narrower
- Wider GOOGLE & Lect. 35, page no. 265 Handouts
- Accurate
- Inaccurate

60. To determine a sample size in estimating population mean, we use the ____ value.

- X
- Z Lect. 36, page no. 271 handouts
- F
- T

61. In the method of moments, how many equations are required for finding two unknown parameters.

- 2 Lect. 34, page no. 258 & Google
- 4
- 1
- 3

62. What is the rule of thumb for a sample size 'n' to be considered large?

- n>30 or n=30 (Lecture no 31 Page no 238 Handouts)
- n=30
- n<30
- None of the above

63. A null hypothesis is generally denoted by:

- H1
- H3
- H2
- H0 (Lecture no 36 Page no 272 Handouts)

64. “To consider every possible value that the parameter might have, and for each value, compute the probability and THAT value of the parameter for which the probability of a given sample is greatest, is chosen as an estimate.” This procedure is known as

- The method of moments
- The method of least square
- The method of maximum likelihood (Lecture no 34 Page no 258 Handouts)
- The method of fractional moments

65. Alpha is the probability of:

- Type-I error
- Committing type-I error
- All of these Lect. 36, page 273 handouts
- Reject H0/H0 is true

66. If P (6.2 < ? < 1.2) = 0.95 then

- **This is wrong statement** Conceptual
- All are correct
- We are 5% confident that true value of the parameter fall in this interval
- We are 95% confident that true value of the parameter fall in this interval

67. The limitation of point estimation is that:

- It is not calculated for small sample
- All of these
- **There is no way of ascertaining how close the point estimate to parameter? (Lecture no 35 Page no 259 Handouts)**
- It is not calculated for non-normal distribution

STA301:QUIZ-2

Question # 1 of 10 (Start time: 01:27:40 AM, 01 February 2021)

The marginal p.d.f. of any continuous variables is obtained by _____ the other variable from the joint p.d.f.

Select the correct option

- ○ Averaging
- ◉ Integrating
- ○ Multiplying
- ○ Dividing

Click to

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STA301:QUIZ-2

Quiz Start

Question # 2 of 10 (Start time: 01:28:07 AM, 01 February 2021)

In a normal distribution how much area lies between $\mu \pm \sigma$

Select the correct option

- 65%
- 68.26%
- 75%
- 80%

Click to Save Ans

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STA301:QUIZ-2

Question # 3 of 10 (Start time: 01:28:29 AM, 01 February 2021)

A correlation coefficient:

Select the correct option

- tells you the direction of the slope of the scattergram.
- is a sort of index of how close the points of a scattergram deviate from the best fitting straight line through those points.
- efficiently summarises some of the information in a scatterplot.
- [x] all of these.

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STA301:QUIZ-2

Question # 4 of 10 (Start time: 01:28:53 AM, 01 February 2021)

The values of moment ratios b1 and b2 of normal distribution are:

Select the correct option

- 0 and 1
- 0 and 2
- 0 and 3
- 0 and 4

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STA301:QUIZ-2

Question # 5 of 10 (Start time: 01:29:15 AM, 01 February 2021)

Which of the following is not a property of a binomial experiment?

Select the correct option

- the experiment consists of a sequence of n identical trials
- each outcome can be referred to as a success or a failure
- the probabilities of the two outcomes can change from one trial to the next
- All of the above

Click to

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Question # 6 of 10 (Start time: 01:29:36 AM, 01 February 2021)

We can obtain the individual probabilities of x and y from the joint probability function/distribution of f(x,y). Such individual probabilities are known as

Select the correct option

- (●) Marginal probabilities
- ( ) Bivariate probabilities
- ( ) Separate probabilities
- ( ) Independent probabilities

Click to Save Answer & M

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Question # 7 of 10 (Start time: 01:29:55 AM, 01 February 2021)

Assume a large distribution in which 6 = 6. What is the standard error of the mean based on samples for which n = I0?

Select the correct option

R

- ○ I67
- ○ 15
- ○ 3
- ● 1.9

Click to Save Answer & Mo

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STA301:QUIZ-2

Question # 8 of 10 (Start time: 01:30:15 AM, 01 February 2021)

In a bivariate probability distribution of X and Y,
If f(1,1)=4/15, g(1)=2/3, h(1)=2/5,
then

Select the correct option

- [x] X and Y are statistically independent
- [ ] X and Y are not statistically independent
- [ ] Both of these
- [ ] None of these

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STA301:QUIZ-2

Question # 9 of 10 (Start time: 01:30:36 AM, 01 February 2021)

In hyper geometric distribution events can be classified in to

Select the correct option

- a single category
- 2 categories
- 3 categories
- 4 categories

Click

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STA301:QUIZ-2

Question # 10 of 10 (Start time: 01:31:01 AM, 01 February 2021)

In binomial distribution, formula of calculating mean is:

Select the correct option

- ○ μ = p + q
- ◉ μ = np
- ○ μ = pq
- ○ μ = qn

Click

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STA301:QUIZ-2

Question # 1 of 10 (Start time: 01:40:10 AM, 01 February 2021)

Which of the following is not a property of a binomial experiment?

Select the correct option

- the experiment consists of a sequence of n identical trials
- each outcome can be referred to as a success or a failure
- the probabilities of the two outcomes can change from one trial to the next
- All of the above

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Question # 2 of 10 (Start time: 01:40:37 AM, 01 February 2021)

Poisson distribution can be used to approximate the hyper geometric distribution when; n <0.05N, n>20 and

Select the correct option

- ○ p>0.05
- ◉ p<0.05
- ○ p>0.10
- ○ p<0.10

Click to Sa

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STA301:QUIZ-2

Question # 3 of 10 (Start time: 01:41:00 AM, 01 February 2021)

The hypergeometric probability distribution has parameters:

Select the correct option

- ○ N, n
- ○ N, P, n
- ◉ N, n, k
- ○ N, N-K, n

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STA301:QUIZ-2

Question # 4 of 10 (Start time: 01:41:23 AM, 01 February 2021)

A random experiment is a one in which we repeat it a large number of times under similiar conditions an it produces:

Select the correct option

- Same results
- Different results
- Independent results
- None of the above

Click

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STA301:QUIZ-2

Question # 5 of 10 (Start time: 01:41:42 AM, 01 February 2021)

Given below is the probability distribution of random variable X. What is the value of "k"

x	0	1	2	3	4
p(x)	0.5	0.2	0.07	k	0.03

Select the correct option

- ○ 0.02
- ○ 0.1
- ● 0.20
- ○ 0.002

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STA301:QUIZ-2

Question # 6 of 10 (Start time: 01:42:02 AM, 01 February 2021)

Poisson distribution is never:

Select the correct option

- ○ Symmetrical
- ○ Positively Skewed
- ◉ Negatively skewed
- ○ Asymmetrical

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STA301:QUIZ-2

Question # 7 of 10 (Start time: 01:42:22 AM, 01 February 2021)

If X and Y are independent, then Cov (X, Y) is equal to:

Select the correct option

- [x] 0
- [ ] 1
- [ ] 0 and 1
- [ ] None

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STA301:QUIZ-2

Question # 8 of 10 (Start time: 01:42:43 AM, 01 February 2021)

If E(x) = 3 then E(4X+3)=.....:

Select the correct option

- (●) 15
- ( ) 12
- ( ) 8
- ( ) 3

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STA301:QUIZ-2

Question # 9 of 10 (Start time: 01:43:01 AM, 01 February 2021)

Which of the following is not a requirement of a binomial distribution?

Select the correct option

- ○ A constant probability of success
- ○ Only two possible outcomes.
- ○ A fixed number of trials.
- ● Equally likely outcomes.

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STA301:QUIZ-2

Question # 10 of 10 (Start time: 01:43:22 AM, 01 February 2021)

Which of the following is the requirement of a uniform distribution?

Select the correct option

- ○ A constant probability of success
- ○ Only two possible outcomes
- ○ A fixed number of trials
- ◉ Equally likely outcomes

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

STA301:QUIZ-3

Question # 1 of 10 (Start time: 10:22:24 PM, 19 February 2021)

If we are testing H0 : mean = 250 against
H1 : mean > 250 (exceeds 250).Then rejection region will be

Select the correct option

- ○ In the center
- ○ on both sides of mean
- ○ on left side of mean
- ◉ on right side of mean

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



STA301 - Statistics and Probability



STA301:QUIZ-3

Quiz Start Time: 10:22 P

Question # 2 of 10 (Start time: 10:22:46 PM, 19 February 2021)

Suppose H0: u 1 < u 2 H1 :u 1 > u 2 and Critical value is = 1.645 If calculated value of Z = 1 .88 then what will be your conclusion?

Select the correct option

Reloa

- ○ Accept Ho
- ◉ Reject Ho
- ○ Impossible to decide
- ○ None of the above

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



STA301 - Statistics and Probability



STA301:QUIZ-3

Question # 3 of 10 (Start time: 10:23:06 PM, 19 February 2021)

The t-distribution can never become narrower than:

Select the correct option

- F-distribution
- standard normal distribution
- Chi-square distribution
- exponential distribution

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَنْ تَشَاءُ وَتَنْزِعُ الْمُلْكَ مِمَّنْ تَشَاءُ وَتُعِزُّ مَنْ تَشَاءُ وَتُذِلُّ مَنْ تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

STA301:QUIZ-3

Question # 4 of 10 (Start time: 10:23:31 PM, 19 February 2021)

t-distribution can be used to find the confidence interval for:

Select the correct option

- Population variance
- Population mean
- Difference between population proportions
- Populating proportion

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَنْ تَشَاءُ وَتَنْزِعُ الْمُلْكَ مِمَّنْ تَشَاءُ وَتُعِزُّ مَنْ تَشَاءُ وَتُذِلُّ مَنْ تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ





STA301:QUIZ-3

Question # 5 of 10 (Start time: 10:23:52 PM, 19 February 2021)

Which one of the following provides the basis for hypothesis testing?

Select the correct option

- ○ Null hypothesis
- ○ Alternative hypothesis
- ○ Critical value
- ◉ Test-statistic

Click t

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

STA301:QUIZ-3

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Question # 6 of 10 (Start time: 10:24:13 PM, 19 February 2021)

For test of hypothesis Ho:μ 1 = μ 2 and H1:μ1 < μ2, the critical region at 0.05 level of significance and n > 30

Select the correct option

- ○ Z less and equal 1.96
- ○ Z> 1.96
- ○ Z>1.645
- ◉ Z<-1.645

Click to Save A

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



STA301 - Statistics and Probability



STA301:QUIZ-3

Qu

Question # 7 of 10 (Start time: 10:24:32 PM, 19 February 2021)

Alpha is the probability of .....

Select the correct option

- (●) Making Type I error
- ( ) Making Type II error
- ( ) Accepting Ho when it is true
- ( ) Rejecting H1 when it is wrong

Click to S

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قلِ اللهم مالِكَ المُلكِ تؤتِي المُلكَ من تشاءُ وتنزِع المُلكَ مِمن تشاءُ وتعِز من تشاءُ وتذِل من تشاءُ بِيدِكَ الخيرُ إنك علىٰ كلِ شيءٍ قدِيرٌ

STA301:QUIZ-3

Question # 8 of 10 (Start time: 10:24:54 PM, 19 February 2021)

The test statistic to test the U1 = U2 (U represent the mean of population)for normal population when n>30.

Select the correct option

- [x] Z-Test
- [ ] T-Test
- [ ] F-Test
- [ ] All of Above

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Quiz Start Ti

Question # 9 of 10 (Start time: 10:25:14 PM, 19 February 2021)

Suppose that the workers of factory B believe that the average income of the workers of factory A exceeds their average income; the null hypothesis will be:

Select the correct option

- ○ MeanA > MeanB
- ○ MeanA ≥ MeanB
- ○ MeanA < MeanB
- ◉ MeanA ≤ MeanB

Click to Save Answe

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Question # 10 of 10 (Start time: 10:25:35 PM, 19 February 2021)

Which of the following statements is NOT true?

Select the correct option

- The t distribution is symmetric about zero
- The t distribution is more spread out than the standard normal distribution
- As the degrees of freedom get smaller, the t-distribution's dispersion gets smaller
- The t distribution is mound-shaped

Click

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Quiz Start

Question # 1 of 10 (Start time: 10:27:09 PM, 19 February 2021)

A _____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

Select the correct option

- Hypothesis
- Lower limit
- Confidence interval
- Probability limit

Click to Save Ans

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

STA301:QUIZ-3

Question # 2 of 10 (Start time: 10:27:28 PM, 19 February 2021)

The proportion of cigarette smokers in Pakistan is greater than 20%, the null hypothesis Ho is:

Select the correct option

- ○ P < 0.20
- ○ P ≠ 0.20
- ● P ≤ 0.20
- ○ P > 0.20

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Question # 3 of 10 (Start time: 10:27:48 PM, 19 February 2021)

A null hypothesis is generally denoted by:

Select the correct option

- [x] H0
- [ ] H1
- [ ] H2
- [ ] H3

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Question # 4 of 10 (Start time: 10:28:09 PM, 19 February 2021)

A deserving player was not selected in the cricket team. It is an example of:

Select the correct option

- (●) Type-I error
- ( ) Type-II error
- ( ) Type-III error
- ( ) Type-IV error

Cli

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَنْ تَشَاءُ وَتَنْزِعُ الْمُلْكَ مِمَّنْ تَشَاءُ وَتُعِزُّ مَنْ تَشَاءُ وَتُذِلُّ مَنْ تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ





STA301:QUIZ-3

Question # 5 of 10 (Start time: 10:28:29 PM, 19 February 2021)

As sample size goes up, what tends to happen to 95% confidence intervals?

Select the correct option

- They become wider
- They become more precise
- They become more narrow
- They become more precise and narrow

Click to S

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Question # 6 of 10 (Start time: 10:28:47 PM, 19 February 2021)

To determine a sample size which test we use .....

Select the correct option

- [x] z
- [ ] t
- [ ] F
- [ ] None of these

Clic

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

STA301:QUIZ-3

Question # 7 of 10 (Start time: 10:29:07 PM, 19 February 2021)

How can we interpret the 90% confidence interval for the mean of the normal population.

Select the correct option

- ○ There are 10% chances of falling true value of the parameter
- ◉ There are 90% chances of falling true value of the parameter
- ○ There are 100% chances of falling true value of the parameter
- ○ All are correct

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Quiz Sta

Question # 8 of 10 (Start time: 10:29:35 PM, 19 February 2021)

For the smaller values of v, the t-distribution will be _____ than the standard normal distribution.

Select the correct option

- less spread out
- more spread out
- same length
- None of these

Click to Save A

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Question # 9 of 10 (Start time: 10:29:59 PM, 19 February 2021)

If we are testing H0 : mean = 50 against
H1 : mean < 50 then the rejection region will be

Select the correct option

- ○ in the center
- ◉ on left side of mean
- ○ on right side of mean
- ○ on both sides

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَنْ تَشَاءُ وَتَنْزِعُ الْمُلْكَ مِمَّنْ تَشَاءُ وَتُعِزُّ مَنْ تَشَاءُ وَتُذِلُّ مَنْ تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Quiz Start Time: 10:27 PM, 19 February 2021

Question # 10 of 10 (Start time: 10:30:17 PM, 19 February 2021) **Total Marks:** 1

If we are testing H0 : P1 < P2(or) P1 - P2 < 0 against H1 : P1 > P2(or) P1 - P2 > 0 and the tabulated value of Z = 1.645 and Z calculated value = 2.63 then what will be the conclusion:
\end{gathered} \]

Select the correct option

Reload Math Equations

- (●) Reject Ho
- ( ) Accept Ho
- ( ) None of the above
- ( ) Impossible to decide

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Question # 1 of 10 (Start time: 10:33:07 PM, 19 February 2021)

In Binomial distribution the sample size is considered to be sufficiently large, if both np and nq are greater than or equal to:

Select the correct option

- 3
- 5
- 10
- 15

Click to

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



STA301 - Statistics and Probability



STA301:QUIZ-3

Quiz Start Time: 10:33 PM, 19 February 202

Question # 2 of 10 (Start time: 10:33:33 PM, 19 February 2021) **Total Marks:**

If the NULL hypothesis says that the average height of Pakistani soldiers exceeds the average height of American soldiers by not more than 3 inches. What will be its ALTERNATIVE hypothesis?

Select the correct option

Reload Math Equations

- (●) P > 3
- ( ) P < 3
- ( ) P = 3
- ( ) none of these

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Question # 3 of 10 (Start time: 10:33:58 PM, 19 February 2021)

The critical region is computed from the value of:

Select the correct option

- [x] Alpha
- [ ] Beta
- [ ] 1-alpha
- [ ] 1-Beta

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ



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STA301:QUIZ-3

Quiz Sta

Question # 4 of 10 (Start time: 10:34:17 PM, 19 February 2021)

To test the hypothesis about the difference of means for large samples, what test statistic will be used?

Select the correct option

- ○ Chi-square
- ○ F
- ● Z
- ○ t

Click to Save An

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَنْ تَشَاءُ وَتَنْزِعُ الْمُلْكَ مِمَّنْ تَشَاءُ وَتُعِزُّ مَنْ تَشَاءُ وَتُذِلُّ مَنْ تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

STA301:QUIZ-3

Question # 5 of 10 (Start time: 10:34:38 PM, 19 February 2021)

When we start hypothesis testing, we always assume that:

Select the correct option

- [x] H0 is true
- [ ] H0 is false
- [ ] H1 is true
- [ ] All are correct

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قلِ اللهم مالِكَ المُلكِ تؤتِي المُلكَ من تشاءُ وتنزِع المُلكَ مِمن تشاءُ وتعِز من تشاءُ وتذِل من تشاءُ بِيدِكَ الخيرُ إنك علىٰ كلِ شيءٍ قدِيرٌ



STA301 - Statistics and Probability



STA301:QUIZ-3

Question # 6 of 10 (Start time: 10:34:56 PM, 19 February 2021)

To test the hypothesis about the difference of means for small samples, test statistics will be used:

Select the correct option

- ○ F-test
- ○ Z-test
- ◉ T-test
- ○ All of these

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

STA301:QUIZ-3

Question # 7 of 10 (Start time: 10:35:18 PM, 19 February 2021)

Ideally, the width of confidence interval should be

Select the correct option

- [x] 0
- [ ] 1
- [ ] 98
- [ ] 100

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

STA301:QUIZ-3

Question # 8 of 10 (Start time: 10:35:40 PM, 19 February 2021)

Conventionally, the probability of making a type-I error is denoted by which of the following symbol?

Select the correct option

- ? (theta)
- a (alpha)
- ß (beta)
- s (sigma)

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

STA301:QUIZ-3

Question # 9 of 10 (Start time: 10:35:58 PM, 19 February 2021)

In a random sample of 500 men from Lahore city, 300 are found to be smokers; the proportion of smokers is equal to:

Select the correct option

- ○ 0.3
- ○ 0.4
- ○ 0.5
- ◉ 0.6

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ بِيَدِكَ الْخَيْرُ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

STA301:QUIZ-3

Question # 10 of 10 (Start time: 10:36:17 PM, 19 February 2021)

If we are testing H0 :P1 - P2 = 120 against H1 : P1 - P2 > 120 (exceeds 120).Then rejection region will be:

Select the correct option

- in Center
- on right side of Z=0
- on left side of Z=0
- on both sides of Z=0

Sonu Ilyas Mughal
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STA301:QUIZ-3 Quiz Start Time: 12:

Question # 1 of 10 (Start time: 12:11:41 AM, 19 February 2021)

When sample size is to be considered large, population standard deviation can be replaced by

Select the correct option

- ○ Population variance
- ○ Sample Variance
- ○ Sample Mean
- ◉ Sample Standard Deviation

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STA301:QUIZ-3 Quiz Sta

Question # 2 of 10 (Start time: 12:12:00 AM, 19 February 2021)

For a particular hypothesis test, Alpha=0.09 and Beta=0.03, what is the value of Type I error?

Select the correct option

- (●) 0.09
- ( ) 0.91
- ( ) 0.03
- ( ) 0.97

RIZ MUGHAL

STA301:QUIZ-3

Question # 3 of 10 (Start time: 12:12:14 AM, 19 February 2021)

Pairing is done only in case of .....observations.

Select the correct option

- (●) Dependent
- ( ) Independent
- ( ) Small
- ( ) Large

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Ti

Question # 4 of 10 (Start time: 12:12:32 AM, 19 February 2021)

How can we interpret the 90% confidence interval for the mean of the normal population.

Select the correct option

- ○ There are 10% chances of falling true value of the parameter
- ◉ There are 90% chances of falling true value of the parameter
- ○ There are 100% chances of falling true value of the parameter
- ○ All are correct

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STA301:QUIZ-3

Question # 5 of 10 (Start time: 12:12:46 AM, 19 February 2021)

Alpha is the probability of:

Select the correct option

- ○ Committing Type I error
- ○ Reject H0/H0 is true
- ○ Type I error
- ◉ All of these

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STA301:QUIZ-3 Quiz Start Time: 12:11 AM, 19 February 20

Question # 6 of 10 (Start time: 12:13:02 AM, 19 February 2021) **Total Marks**

Suppose H0: p =0 H1:p < 0 and Critical value is z = - 2.33. If calculate the value of Z = - 2.60,then what will be your conclusion?

Select the correct option

Reload Math Equation

- ○ Accept Ho
- ◉ Reject Ho
- ○ Impossible to decide
- ○ None of the above

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Time: 12:11 AM, 19 February 2021

Question # 7 of 10 (Start time: 12:13:19 AM, 19 February 2021) **Total Marks:** 1

Suppose that the workers of factory B believe that the average income of the workers of factory A exceeds their average income; the null hypothesis will be:

Select the correct option

- ○ MeanA > MeanB
- ○ MeanA ≥ MeanB
- ○ MeanA < MeanB
- ◉ MeanA ≤ MeanB

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STA301:QUIZ-3 Quiz

Question # 8 of 10 (Start time: 12:13:36 AM, 19 February 2021)

If a significance level of 5% is used rather than 1%, the null hypothesis is:

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Select the correct option

- ○ Less likely to be rejected
- ○ Just as likely to be rejected
- ● More likely to be rejected
- ○ None of these

STA301:QUIZ-3

Quiz Start Time: 12:11 AM, 19 February 2021

Question # 9 of 10 (Start time: 12:13:51 AM, 19 February 2021)

Total Marks: 1

If the NULL hypothesis says that the average height of Pakistani soldiers exceeds the average height of American soldiers by not more than 3 inches. What will be its ALTERNATIVE hypothesis?

Select the correct option

Reload Math Equations

- [x] P > 3
- [ ] P < 3
- [ ] P = 3
- [ ] none of these

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Time: 12:11 AM, 19 Februa

Question # 10 of 10 (Start time: 12:14:06 AM, 19 February 2021) Total M

Suppose H0: u 1 < u 2 H1 :u 1 > u 2 and Critical value is = 1.645 If calculated value of Z = 1 .88 then what will be your conclusion?

Select the correct option

Reload Math Equ

○	Accept Ho
◉	Reject Ho
○	Impossible to dec
○	None of the above

RIZ MUGHAL

2nd account

STA301:QUIZ-3

Question # 1 of 10 (Start time: 12:15:35 AM, 19 February 2021)

The term 1- β is called

Select the correct option

- ○ Level of the test
- ◉ Power of the test
- ○ Size of the test
- ○ Critical region

RIZ MUGHAL

STA301:QUIZ-3

Question # 2 of 10 (Start time: 12:15:53 AM, 19 February 2021)

The t-distribution can never become narrower than:

Select the correct option

- ○ F-distribution
- ◉ standard normal distribution
- ○ Chi-square distribution
- ○ exponential distribution

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Time: 12:15 AM, 19 Febru

Question # 3 of 10 (Start time: 12:16:11 AM, 19 February 2021) Total

When examining group difference where no direction of the difference is specified, which of the following is used?

RIZ MUGHAL

Select the correct option

- ○ One-tailed test
- ◉ Two-tailed test
- ○ Directional hypothesis
- ○ Difference of modes test

STA301:QUIZ-3 Quiz Sta

Question # 4 of 10 (Start time: 12:16:26 AM, 19 February 2021)

If we are testing H0 : mean = 50 against
H1 : mean < 50 then the rejection region will be

Select the correct option

- ○ in the center
- ◉ on left side of mean
- ○ on right side of mean
- ○ on both sides

RIZ MUGHAL

STA301:QUIZ-3 Qui

Question # 5 of 10 (Start time: 12:16:41 AM, 19 February 2021)

The level of significance is also called:

Select the correct option

- ○ Power of test
- ○ Type II error
- ◉ Type I error
- ○ Acceptence region

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Time: 12:15 AM, 19 February 2021

Question # 6 of 10 (Start time: 12:16:54 AM, 19 February 2021) **Total Marks: 1**

A random sample of n=25 values gives sample mean 83. Can this sample be regarded as drawn from a normal population with μ= 80 and s= 7? In this question the alternative hypothesis will be:

Select the correct option

- ○ H1: μ = 80
- ◉ H1: μ ≠ 80
- ○ H1: μ > 80
- ○ H1: μ <80

RIZ MUGHAL

STA301:QUIZ-3

Question # 7 of 10 (Start time: 12:17:10 AM, 19 February 2021)

The t-test for independent samples assumes which of the following?

Select the correct option

- (●) Scores are independent
- ( ) Scores are dependent
- ( ) Groups have equal means
- ( ) Groups have equal medians

RIZ MUGHAL

STA301:QUIZ-3 Quiz St

Question # 8 of 10 (Start time: 12:17:23 AM, 19 February 2021)

The critical region is computed from the value of:

Select the correct option

- (●) Alpha
- ( ) Beta
- ( ) 1-alpha
- ( ) 1-Beta

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Time: 12:15 AM, 19 Febru

Question # 9 of 10 (Start time: 12:17:40 AM, 19 February 2021) Tota

Suppose H0: u 1 < u 2 H1 :u 1 > u 2 and Critical value is = 1.645 If calculated value of Z = 1 .88 then what will be your conclusion?

Select the correct option

Reload Math Eq

○	Accept Ho
◉	Reject Ho
○	Impossible to decide
○	None of the above

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Time: 12:15 AM.

Question # 10 of 10 (Start time: 12:17:58 AM. 19 February 2021)

For a sample p1 = 0.2, n1= 100 and for second sample p2= 0.5 and n2= 200, the pooled estimator pc is:

RIZ MUGHAL

Select the correct option

- ○ 0.30
- ○ 0.45
- ◉ 0.40
- ○ 0.35

3rd account

STA301:QUIZ-3 Quiz

Question # 1 of 10 (Start time: 10:49:16 AM, 19 February 2021)

The location of the critical region depends upon:

Select the correct option

- ○ Null hypothesis
- ◉ Alternative hypothesis
- ○ Value of alpha
- ○ Value of test-statistic

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Time: 10

Question # 2 of 10 (Start time: 10:49:34 AM, 19 February 2021)

To test the hypothesis about the difference of means for small samples, which test statistic will be used?

Select the correct option

- ○ Z-test
- ◉ t- test
- ○ F-test
- ○ All of these

RIZ MUGHAL

STA301:QUIZ-3

Question # 3 of 10 (Start time: 10:49:48 AM, 19 February 2021)

The level of significance is also called:

Select the correct option

- ○ Power of test
- ○ Type II error
- ◉ Type I error
- ○ Acceptence region

RIZ MUGHAL

STA301:QUIZ-3

Question # 4 of 10 (Start time: 10:50:01 AM, 19 February 2021)

Pairing is done only in case of .....observations.

Select the correct option

- (●) Dependent
- ( ) Independent
- ( ) Small
- ( ) Large

RIZ MUGHAL

STA301:QUIZ-3

Question # 5 of 10 (Start time: 10:50:14 AM, 19 February 2021)

Critical region is computed with the value of .....

Select the correct option

- (●) α
- ( ) ß
- ( ) 1-α
- ( ) 1- ß

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Time: 10:49 AM, 19 February 2

Question # 6 of 10 (Start time: 10:50:28 AM, 19 February 2021) Total Mar

Suppose H0: u 1= u2 and H1:u 1 < u 2 and Critical value is z= - 1.645. If calculated value of Z= - 2.08,then what will be your conclusion?

Select the correct option

Reload Math Equatic

- ○ Accept Ho
- ◉ Reject Ho
- ○ Impossible to decide
- ○ None of the above

RIZ MUGHAL

STA301:QUIZ-3 Q

Question # 7 of 10 (Start time: 10:50:44 AM, 19 February 2021)

The t-test for independent samples assumes which of the following?

Select the correct option

- (●) Scores are independent
- ( ) Scores are dependent
- ( ) Groups have equal means
- ( ) Groups have equal medians

RIZ MUGHAL

STA301:QUIZ-3

Quiz Start Time: 10:49 AM, 19 February 2021

Question # 8 of 10 (Start time: 10:51:02 AM, 19 February 2021) **Total Marks:** 1

A random sample of n=25 values gives sample mean 83. Can this sample be regarded as drawn from a normal population with μ= 80 and s= 7? In this question the alternative hypothesis will be:

Select the correct option

- ○ H1: μ = 80
- ◉ H1: μ ≠ 80
- ○ H1: μ > 80
- ○ H1: μ <80

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Tim

Question # 9 of 10 (Start time: 10:51:18 AM, 19 February 2021)

Mean of t-distribution will not exist when d.f(degree of freedom) will equal to:

Select the correct option

- (●) 1
- ( ) 2
- ( ) 3
- ( ) 4

RIZ MUGHAL

STA301:QUIZ-3 Quiz Start Time: 10:49 A

Question # 10 of 10 (Start time: 10:51:34 AM, 19 February 2021)

Which of the following is not a criterion for a good problem situation?

Select the correct option

- It must be authentic.
- It must be clearly structured and defined.
- It must be appropriate to the student's level of development.
- It must be such that its solution will be benefited by group effort.

RIZ MUGHAL

STA301:QUIZ-2 Quiz Start Ti

Question # 1 of 10 (Start time: 07:48:22 AM, 01 February 2021)

Which of the following statements is not an example of a continuous random variable?

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Select the correct option

- ○ The weight gain in pounds per month for a calf
- ○ The price for cheese cake in New York Style cheese cake
- ○ The time it takes you to finish this statistics test
- ◉ The time to failure for an electronic system

STA301:QUIZ-2 Quiz Start Time: 07:48 AM, 01 February 2021

Question # 2 of 10 (Start time: 07:48:48 AM, 01 February 2021) **Total Marks:** 1

Consider a large population with a mean of 160 and a standard deviation of 20. A random sample of size 64 is taken from this population. What is the standard deviation of the sample mean?

Select the correct option

- 3.125
- 2.500
- 1.654
- 3.568

RIZ MUGHAL

STA301:QUIZ-2 Quiz Start Time: 07:

Question # 3 of 10 (Start time: 07:49:05 AM, 01 February 2021)

Suppose 'X' is equal to the number of heads, when you throw a coin three times. The possible values of X are:

Select the correct option

- ○ 1,2,3
- ◉ 0,1,2,3
- ○ 0,1,2,3,.....
- ○ 1,2,3,.....

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STA301:QUIZ-2

Question # 4 of 10 (Start time: 07:49:25 AM, 01 February 2021)

If we have f(x) = 4x, 0≤x≤1, then f(x) is a:

Select the correct option

- ○ Continuous random variable
- ○ Distribution function
- ◉ Probability density function
- ○ Probability distribution

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STA301:QUIZ-2

Quiz Start Time: 07:48 AM, 01 February 2021

Question # 5 of 10 (Start time: 07:49:40 AM, 01 February 2021) **Total Marks: 1**

The sum of the probabilities of bivariate probabilities function f(X, Y) is ONE then what will be the value of sum of its marginal probability function f(x)?

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Select the correct option

- (●) Will be one
- ( ) May and may not be one
- ( ) Will be the average of two variables
- ( ) Depends on the Y variable

STA301:QUIZ-2

Question # 6 of 10 (Start time: 07:49:58 AM, 01 February 2021)

If the second moment ratio is less than 3 the distribution will be:

RIZ MUGHAL

Select the correct option

- ○ Mesokurtic
- ○ Leptokurtic
- ◉ Platykurtic
- ○ None of these

STA301:QUIZ-2

Question # 7 of 10 (Start time: 07:50:13 AM, 01 February 2021)

E(10X +3) =_____

RIZ MUGHAL

Select the correct option

- ○ 10 E(X)
- ○ E(X)+3
- ◉ 10 E(X)+3
- ○ 100E(X)

STA301:QUIZ-2

Question # 8 of 10 (Start time: 07:50:29 AM, 01 February 2021)

Which of the probability distributions has three parameters?

RIZ MUGHAL

Select the correct option

- ○ Binomial distribution
- ○ Normal distribution
- ◉ Hypergeometric distribution
- ○ Poisson distribution

STA301:QUIZ-2 Quiz S

Question # 9 of 10 (Start time: 07:50:42 AM, 01 February 2021)

Let X and Y are TWO random variables. Which property of expectation is true?

Select the correct option

- E(X + Y) = 2 * {E(X) + E(Y)}
- E(X + Y) = E(X) + E(Y) - E(XY)
- E(X + Y) = 2 E(X + Y)
- E(X + Y) = E(X) + E(Y)

RIZ MUGHAL

STA301:QUIZ-2

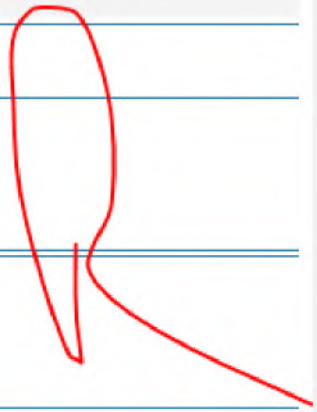
Question # 10 of 10 (Start time: 07:50:57 AM, 01 February 2021)

If f(x) = 1/10, x = 10, then E(X) is:

Select the correct option

- 0
- 1
- -1
- 6/8

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2nd account

STA301:QUIZ-2 Quiz Start Time:

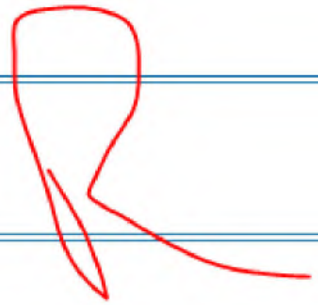
Question # 1 of 10 (Start time: 07:52:46 AM, 01 February 2021)

Using the normal approximation to the binomial distribution with n= 3 and p= 0.0571 the value of mean is:

Select the correct option

- (●) 0.1713
- ( ) 0.2132
- ( ) 0.5133
- ( ) 0.1923

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STA301:QUIZ-2

Question # 2 of 10 (Start time: 07:53:13 AM, 01 February 2021)

When a coin is tossed 3 times, the probability of getting 3 or less tails is

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Select the correct option

- ○ 1/2
- ○ 0
- ◉ 1
- ○ 3/5

STA301:QUIZ-2

Question # 3 of 10 (Start time: 07:53:26 AM, 01 February 2021)

Which of the probability distributions has three parameters?

Select the correct option

- ○ Binomial distribution
- ○ Normal distribution
- ◉ Hypergeometric distribution
- ○ Poisson distribution

RIZ MUGHAL

STA301:QUIZ-2 Quiz Start Time: 07:5

Question # 4 of 10 (Start time: 07:53:47 AM, 01 February 2021)

In a normal distribution how much area lies between $\mu \pm 3\sigma$

Select the correct option

Re

- 70%
- 89%
- 68%
- 99.73%

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STA301:QUIZ-2 Quiz Start Time: 0

Question # 5 of 10 (Start time: 07:54:03 AM, 01 February 2021)

In a normal distribution how much area lies between $\mu \pm 2\sigma$

Select the correct option

- 94%
- 75%
- 95.45%
- 85%

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STA301:QUIZ-2

Question # 6 of 10 (Start time: 07:54:18 AM, 01 February 2021)

If E(x) = 2.5 then E(2X+0)=.....:

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Select the correct option

- 2.5
- 5
- 0
- 3.5

STA301:QUIZ-2 Quiz Start Time: 07:52 AM, 01 February 2

Question # 7 of 10 (Start time: 07:54:32 AM, 01 February 2021) **Total Mar**

We can apply binomial distribution formula, when we are drawing a sample from finite population without replacement and the sample size "n" is not more than .....% of the population size N.

Select the correct option

- 10%
- 8%
- 5%
- 1%

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STA301:QUIZ-2

Question # 8 of 10 (Start time: 07:54:49 AM, 01 February 2021)

Normal approximation to the binomial distribution is used when:

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Select the correct option

- np>5
- nq>5
- Both of the above
- None of the above

STA301:QUIZ-2

Quiz Start Time: 07:52 AM, 01 February 2021

Question # 9 of 10 (Start time: 07:55:05 AM, 01 February 2021) **Total Marks:** 1

We can obtain the individual probabilities of x and y from the joint probability function/distribution of f(x,y). Such individual probabilities are known as

Select the correct option

- (●) Marginal probabilities
- ( ) Bivariate probabilities
- ( ) Separate probabilities
- ( ) Independent probabilities

RIZ MUGHAL

STA301:QUIZ-2

Question # 10 of 10 (Start time: 07:55:25 AM, 01 February 2021)

If E(x) = 1.5 then E(2X+3)=.....:

Select the correct option

- ○ 3
- ○ 2
- ○ 1.5
- ◉ 6

RIZ MUGHAL

3rd account

STA301:QUIZ-2 Quiz Start Time: 07:59 AM, 01 February 2

Question # 1 of 10 (Start time: 07:59:06 AM, 01 February 2021) **Total Mark**

If the sampled population has a normal distribution, when is the sampling distribution of the sample mean also a normal distribution?

Select the correct option

- ○ Only when the sample size is at least 100
- ○ Only when the sample size is smaller than 30
- ◉ Always
- ○ Only when the sample size is at least 30

RIZ MUGHAL

STA301:QUIZ-2 Quiz

Question # 2 of 10 (Start time: 07:59:51 AM, 01 February 2021)

Which of the following statements is not an example of a continuous random variable?

Select the correct option

- ○ The weight gain in pounds per month for a
- ○ The price for cheese cake in New York Style cheese cake
- ○ The time it takes you to finish this statistics test
- ◉ The time to failure for an electronic system

RIZ MUGHAL

STA301:QUIZ-2 Quiz

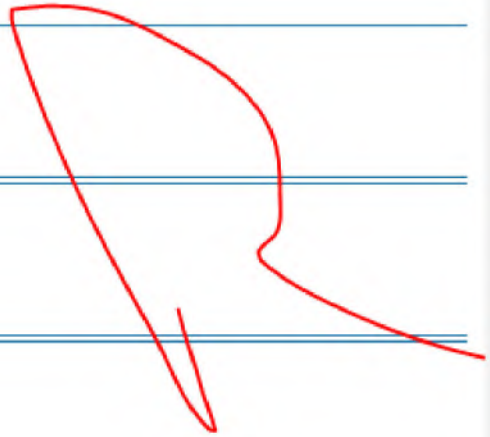
Question # 3 of 10 (Start time: 08:00:13 AM, 01 February 2021)

The probability mass function p(x) of a discrete random variable X is: P(0)=0.10, P(1)=0.15, P(2)=0.45, P(3)=0.05, and P(4)=0.25.
What is the value of cumulative distribution function for x=2

Select the correct option

- ○ 0.3
- ○ 0.5
- ◉ 0.7
- ○ 0.5

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STA301:QUIZ-2

Question # 4 of 10 (Start time: 08:00:30 AM, 01 February 2021)

In a binomial experiment successive trials are:

Select the correct option

- ○ Dependent
- ◉ Independent
- ○ Both
- ○ None

RIZ MUGHAL

STA301:QUIZ-2

Question # 5 of 10 (Start time: 08:00:48 AM, 01 February 2021)

Match the binomial probability P(x< 23) with the correct statement.

Select the correct option

- ○ P(there are at most 23 successes)
- ◉ P (there are fewer than 23 successes)
- ○ P (there are more than 23 successes)
- ○ P (there are at least 23 successes)

RIZ MUGHAL

STA301:QUIZ-2

Question # 6 of 10 (Start time: 08:01:02 AM, 01 February 2021)

The two mutually exclusive outcomes in a Bernoulli trial are usually called:

Select the correct option

- (●) Success and failure
- ( ) Variable and constant
- ( ) Mean and variance
- ( ) With and without replacement e trials are independent

RIZ MUGHAL

STA301:QUIZ-2

Question # 7 of 10 (Start time: 08:01:16 AM, 01 February 2021)

If the first moment ratio is equal to 0, then the distribution will be:

Select the correct option

- ○ Positively Skewed
- ○ Leptokurtic
- ○ Negatively Skewed
- ◉ Symmetrical

RIZ MUGHAL

STA301:QUIZ-2 Quiz

Question # 8 of 10 (Start time: 08:01:33 AM, 01 February 2021)

What does it mean when a data set has a standard deviation equal to zero:

Select the correct option

- ○ Mean = median = mode
- ◉ All of the values in the data are same
- ○ The mean of the data is also zero
- ○ All values of the data appear with the same frequency

RIZ MUGHAL

STA301:QUIZ-2

Question # 9 of 10 (Start time: 08:01:49 AM, 01 February 2021)

The mean deviation of the normal distribution is approximately:

Select the correct option

- 7/8 of the S.D
- 4/5 of the S.D
- 3/4 of the S.D
- 1/2 of the S.D

RIZ MUGHAL

STA301:QUIZ-2 Quiz Start Time: 07:59 AM, 01 February 202

Question # 10 of 10 (Start time: 08:02:05 AM, 01 February 2021) **Total Marks:**

A sales firm receives different calls from clients. If someone says, today it will receive "at most 3 calls". How many calls will be received?

Select the correct option

- 1, 2, 3
- 0, 1, 2, 3
- 3, 4, 5, ...
- 0, 1, 2, 3, 5, ...

RIZ MUGHAL



Final Term Papers Solved MCQS with Reference

1. The parameter of the chi-square distribution is……

- o **v** **PG # 306 (Lec # 41)**
- o v – 1
- o v – 2
- o v –p

2. The value of x^2 can never be :

- o Zero
- o Less than 1
- o Greater than 1
- o **Negative**

3. Analysis of variance is a procedure that enables us to test the equality of several:

- o Variances
- o **Means** **PG # 320 (Lec # 43)**
- o Proportions
- o Groups

4. When two coins are tossed the probability of at most one head is:

- o 1/4
- o 2/4
- o **3/4**
- o 4/4

5. The critical region for $H_1 : \mu < \mu_0$ *when* $\alpha = 0.05$ is

- $z > z_{0.05}$ **PG # 281 (Lec # 37)**
- $|z| > z_{0.01}$
- $z < -z_{0.05}$
- $|z| > z_{0.05}$

6. A discrete probability function f(x) is always:

- **Non -negative**
- Negative
- One
- Zero

Note: A discrete distribution describes the probability of occurrence of each value of a discrete random variable. A discrete random variable is a random variable that has countable values, such as a list of non-negative integers.

7. When f(x) is When f(x) is continuous probability function for, then P(X = 1) is:

- 1
- ∞
- $-\infty$
- **0** **PG # 188 (Lec # 24)**

8. A random variable can be generated:

- Manually
- Mechanically
- **Manually & Mechanically** **PG # 167 (Lec # 22)**
- Mathematically

Note: Such a numerical quantity whose value is determined by the outcome of a random experiment is called a random variable. Random numbers are the numbers obtained by some random process (manually or mechanically).

9. The curve of the F-distribution depends upon:

- **Degree of freedom** **PG # 312 (Lec # 42)**
- Standard deviation
- Mean
- Variance

10. The mode value from raw data can be obtained by the help of

- **Dot plot** **PG # 53 (Lec #6)**
- Stem and leaf plot
- Bar chart
- None of these

11. In normal distribution, the quartile deviation Q.D =…….:

- 0.5σ
- 0.75σ
- 0.7979σ
- 0.6745σ

Note: The quartiles are given by:
Lower quartile $Q_1 = \mu - 0.6745\sigma$ and **Upper quartile** $Q_3 = \mu + 0.6745\sigma$

12. A randomly selected sample of 400 students at university was asked whether or not they will participate in politics. Forty-six percent of the 400 student surveyed answered " yes". Which one of the following statement about number 46% is correct?

- **It is a sample statistic.**
- It is a population parameter.
- It is a margin of error.
- It is a standard error.

Note: In statistics a data sample is a set of data collected and/or selected from a statistical population by a defined procedure. The elements of a sample are known as sample points, sampling units or observations.

13. A student solved 25 questions from first 50 questions of a book. The probability that he will solve the remaining all questions is:

- 0.25
- **0.5**
- 1
- 0

14. The average which is defined as the reciprocal of the arithmetic mean of the reciprocals of the values is called:

- Geometric Mean
- **Harmonic Mean** **PG # 77 (Lec # 9)**
- Mode
- Median

15. For the independent events A and B if P (A) = 0.25, P (B) =0.40 then P $(A \cap B)$ =……

- 0.65
- **0.1** **PG # 162 (Lec # 21)**
- 0.50
- 0.15

16. The total number of samples when sampling is done with replacement is equal to:

- N^n **PG # 237 (Lec # 31)**
- C_n^N
- $\frac{N-n}{N-1}$
- 1

17. Which of the following is correct option to find P(X +Y < 1) ?

- f(0 , 0) + f(0,1) + f(1,2)
- f(2,0) + f(0, 1) + f(1, 0)
- f(0, 0) +f (1, 1) + f(1, 0)
- **f (0,0)+ f (0,1) + f (1,0)** **PG # 197 (Lec # 26)**

18. The conditional probability function $f(x|1) =$ _____

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$ **PG # 198 (Lec # 26)**
- $\frac{f(x,1)}{h(x)}$

19. In interval estimation, we always get:

- **Range of values** **PG # 264 (Lec # 34)**
- Zero
- Single value
- Two values

20. Which of the following is an unbiased estimator of population σ^2 ?

- o $\frac{\sum(x-\bar{x})^2}{n}$
- o $\frac{\sum(x-\bar{x})^2}{n-2}$
- o $\frac{\sum(x-\bar{x})^2}{n(n-1)}$
- o $\frac{\sum(x-\bar{x})^2}{(n-1)}$ **PG # 267 (Lec # 34)**

21. How many parameter(s) are in t-distribution?

- o 0
- o **1** **PG # 293 (Lec # 39)**
- o 2
- o 3

22. A deserving player is not selected in the team is an example of:

- o **Type I error** **PG # 278 (Lec # 36)**
- o Type II error
- o Correct decision
- o No information regarding this

23. After an interval is constructed, then what is the probability of occurrence of the parameter in it?

- o Zero
- o One
- o Fifty
- o **Either one or zero** **PG # 268**

24. If $\bar{X}$ is the mean of the n observations, then which test statistic will be used to calculate the confidence limits of the population variance σ^2 ?

- o Z-statistic
- o T-statistic
- o x^2 **-statistics**
- o F-statistics

25. If mean of x^2 distribution is k then variance will be:

- k^2
- $2k$ Click Here For Reference
- $1/k$
- K

26. What is the graphical shape of the chi-square distribution?

- **Positively skewed** **PG # 307 (Lec # 41)**
- Negatively skewed
- Uniformly distributed
- Normally distributed

27. What factor determines the shape of the t-distribution?

- **Degree of freedom** **PG # 293**
- Critical value
- Frequency of data
- Probability

28. If X and Y are random variables, then $E(X-Y)$ is equal to:

- $E(X)+E(Y)$
- $E(X)-E(Y)$ **PG # 202 (Lec # 26)**
- $X-E(Y)$
- $E(X)-Y$

29. In the regression line Y= a + bX, the non-random variable is:

- Y
- X
- Both X and Y
- **Neither X nor Y**

Note: X and Y are two random variables. We assume that Y depends on X. i.e., when variable X takes a specific value, we expect a response in the random variable Y. In other words, the value taken by X influences the value of Y. So X is the independent variable and Y is the dependent variable.

30. The parameters of the binomial distribution b(x; n, p) are:

- x&n
- x&p
- **n&p** **PG # 212**
- x, n & p

Note: Because the pmf (probability mass function) of a binomial random variable X depends on the two parameters n and p, we denote the pmf by b(x; n, p).

31. Which of the following is true for the binomial distribution b(x; n, p):

- **mean > variance** **Click Here For Reference**
- mean < variance
- mean=variance
- mean= standard deviation

32. Hypergeometric probability distribution has :

- (n, k) parameter
- (N) parameter
- (N, n, N-k) parameter
- **(N, n, k) parameter** **PG # 219 (Lec # 28)**

33. The variance of the hypergeometric probability distribution is:

- $\sigma^2 = n\frac{k}{N}$
- $\sigma^2 = n\frac{k}{N}\frac{N-k}{N}\frac{N-n}{N-1}$, **PG # 221 (Lec # 29)**
- $\sigma^2 = npq$
- $\sigma^2 = N\frac{n}{K}$

34. Tabulation is the process of arranging data into:

- Different classes
- Rows
- Columns
- **Rows and Columns**

Note: Tabulation is a systematic presentation of numerical data in row and columns. Tabulation of classified data makes it more intelligible and fit for statistical analysis.

35. A simple bar chart consists of:

- Vertical bars
- Horizontal bars
- **Vertical or Horizontal bars** **PG # 24 (Lec # 3)**
- Multiple bars

36. Which one is the formula to calculate the approximate value of class interval?

- Minimum value/Range
- Maximum value/No. of classes
- **No. of classes/Range** **PG # 29 (Lec # 4)**
- Range/ No. of classes

37. Which one of the following is equal to the 2nd quartile:

- P_{33}
- D_3
- **Median** **PG # 69 (Lec # 8)**
- Mode

38. For the independent events A and B if P (A) = 0.15, P (B) = 0.50 then P $(A \cap B)$ =…….

- 0.50
- **0.075** **PG # 162**
- 0.125
- 0.060

39. Which one of the following measure is not based on all the observations?

- Arithmetic Mean
- Geometric Mean
- Harmonic Mean
- **Mode**

Note: Mode is that value of the variable which occurs or repeats itself maximum number of item.

40. When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

- Mutually exclusive
- **Equally likely** **PG # 147 (Lec # 17)**
- Not mutually exclusive
- Exhaustive

41. An expected value of a random variable is equal to:

- Variance
- **Mean** **PG # 191(Lec # 25)**
- Standard deviation
- Covariance

42. The sum of deviations is zero, when deviations are taken from:

- **Mean**
- Median
- Mode
- H.M

43. The distribution function F(x) is equal to

- $P(X = x)$
- $P(X \leq x)$ **PG # 174 (Lec # 23)**
- $P(X \geq x)$
- $P(X > x)$

44. In a one-way ANOVA:

- The interaction term has (c - l)(n - 1) degrees of freedom
- An interaction term is given
- **An interaction effect can be tested** **Click Here For Reference**
- There is no interaction term

45. The degrees of freedom for a t-test with sample size 'n' is:

- **n-1** **PG # 341 (Lec # 45)**
- n+1
- n-2
- n+2

46. Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the null hypothesis is:

- $H_0 : \mu = 100$ **PG # 278 (Lec # 36)**
- $H_0 : \mu > 100$
- $H_0 : \mu < 100$
- $H_0 : \mu \neq 100$

47. When c is a constant, then E(c) is:

- 0
- 1
- **c** **PG # 180 (Lec # 23)**
- –c

48. The combined distribution of more than two random variables is:

- Univariate distribution
- **Joint distribution** **PG # 194 (Lec # 25)**
- Marginal distribution
- Bivariate distribution

49. The test statistic used in analysis of variance procedure follow the :

- x^2 -distribution
- T- distribution
- Z-distribution
- **F- distribution** **Click Here For Reference**

50. In normal distribution β_2 =…….:

- 1
- 2
- **3** **PG # 119 (Lec # 14)**
- 0

51. In a symmetrical distribution, the coefficient of skewness is equal to :

- –1
- +1
- **0**
- 2

Note: If the coefficient of skewness is equal to zero, the shape of the distribution is symmetric.

52. Which measure of dispersion is used to compare the variation of two data sets?

- **Coefficient of variation** **PG # 92 (Lec # 11)**
- Coefficient of comparison
- Mean deviation
- Standard deviation

53. If S.D(X) = 5 then S.D $\left(\frac{2X+5}{2}\right)$ = _____

- 5
- 10
- 15
- **7.5**

54. The deviation of a distribution from symmetry is called:

- Kurtosis
- **Skewness**
- Dispersion
- Flatness

55. Which one is the correct formula to find the desired sample size?

- $n = \left(\frac{Z_{\alpha/2}\sigma}{e}\right)^2$ **PG # 276 (Lec # 36)**
- $n = \left(\frac{Z_{\alpha/2}\sqrt{\sigma}}{e}\right)^2$
- $n = \left(\frac{Z_{\alpha/2}\bar{X}}{e}\right)^2$
- $n = \frac{Z_{\alpha/2}\sigma}{e}$

56. A judge can acquit a guilty person is the example of:

- Type I error
- Type II error
- **Correct decision** **PG # 278**
- No information regarding this

57. Ideally, the width of confidence interval should be:

- **0** **PG # 270**
- 1
- 99
- 100

58. If the sampling distribution of $\bar{X}$ is normal, we would expect 99% of the sample means to be within the interval:

- o $\mu_x \pm 2\sigma_x$
- o $\mu_x \pm 1.96\sigma_x$
- o $\mu_x \pm 2.58\sigma_x$
- o $\mu_x \pm \sigma_x$

59. Mean of the F - distribution is possible only, when

- o $v_1 > 2$
- o $v_2 > 2$ **PG # 312 (Lec # 42)**
- o $v_1 < 2$
- o $v_2 < 2$

60. In Statistics, we have MSE which is abbreviation of:

- o **Mean square error** **PG # 330 (Lec # 44)**
- o Measured square error
- o Medical screening exam
- o Major sampling error

61. As the degree of freedom increases, the t-distribution tends to coincide with:

- o Binomial distribute
- o Uniform distribution
- o Hypergeometric distribution
- o **Normal distribution** **PG # 293 (Lec # 39)**

62. If X and Y are independent variables, then E (XY) is:

- o E(XX)
- o **E(X).E(Y)**
- o X.E(Y)
- o Y.E(X)

63. What are the number of ways in which four books can be arranged on a shelf?

- o 4
- o 6
- o 12
- o **24** **PG # 141 (Lec # 17)**

64. When f(x) is continuous probability function for 1< X < 5, then P (X < 1) is:

- 0
- **0.25**
- 0.5
- 1

65. For any two estimators T1 and T2, if VAR (T1) < VAR(T2), then T1 is:

- Unbaised
- Sufficient
- **Efficient** **PG # 261 (Lec # 34)**
- Consistent

66. If an estimator gets closer to the population parameter by increasing sample size then it is known as:

- **Consistent estimator** **PG # 260 (Lec # 33)**
- Sufficient estimator
- Efficient estimator
- Unbiased estimator

67. Which of the following comes first to make frequency distribution.

- Number of Groups
- Class interval
- **Range** **PG # 28 (Lec # 4)**
- Tally marks

68. What curve shape would you expect for the distribution of death rates of population of all age groups?

- Symmetrical curve
- Skewed to the right
- Skewed to the left
- **U shape curve** **PG # 41 (Lec # 5)**

69. Which one is the measure of central tendency:

- Variation of the distribution
- **Average of the distribution** **PG # 51 (Lec # 6)**
- Scatterness of the distribution
- Dispersion of the distribution

70. The mean of the F–distribution is:

- o $\frac{v_1}{v_1-2}\ for v_1 > 2$
- o $\frac{v_2}{v_2-2}\ for v_2 > 2$ **PG # 312 (Lec # 42)**
- o $\frac{v_1}{v_1-2}\ for v_1 \geq 2$
- o $\frac{v_2}{v_2-2}\ for v_1 \leq 2$

71. The F-distribution always ranges from:

- o 0 to 1
- o 0 to $-\infty$
- o $-\infty$ to $+\infty$
- o **0 to $+\infty$** **PG # 312 (Lec # 42)**

72. Which of the following is a characteristics of the normal distribution:

- o It is a skewed distribution
- o **It is bell-shaped** **PG # 226**
- o It is not asymptotic
- o It is leptokurtic

Note: What are the characteristics of the normal distribution

- Unimodal
- Symmetrical
- "Bell-shaped"
- The left and right tails continue to infinity without touching the X-axis
- The mean is the population mean (mew)
- The standard deviation is the population standard deviation (little sigma)

73. For the given poisson distribution $P(X=1)\ \frac{e^{-0.135}0.135^1}{1!}$ the mean value is:

- o $e^{-0.135}$
- o -0.135
- o 0.135 **PG # 223 (Lec # 29)**
- o 1

74. A good way to get a small standard error is to use a _____

- Repeated sampling
- Small sample
- **Large sample**
- Large population

75. The difference between the largest and the smallest data values is called the

- Variance
- Interquartile range
- **Range**
- Coefficient of variation

76. Which is appropriate average for finding the average speed of a car:

- Mean
- Geometric mean
- **Harmonic mean** **PG # 78 and 79 (Lec # 9)**
- Weighted mean

77. Which one is the formula of mid range:

- $x_m - x_0$
- $x_0 - x_m$
- $\frac{x_0 - x_m}{2}$
- $\frac{x_0 + x_m}{2}$ **PG # 80 (Lec # 9)**

78. Which one of the following is a meso-kurtic curve?

- Negatively skewed
- Positively skewed
- J-shaped
- **Normal** **PG # 114 (Lec # 14)**

79. If you draw all possible samples from some population, calculate the mean for each of the sample and construct the probability distribution of the sample means, what would you have?

- A population distribution
- Logical sample distribution
- **A sampling distribution** **PG # 242 (Lec # 31)**
- A parameter distribution

80. By definition f(y | x)= _____

- o $f(y)$
- o $f(x, y)$
- o $\frac{f(x, y)}{h(x)}$ **PG # 198 (Lec # 26)**
- o $\frac{f(x, y)}{h(y)}$

81. For degree of freedom $v > 2$ the variance of t-distribution is always:

- o Greater than zero
- o Less than one
- o Equal to one
- o **Greater than one** **PG # 294 (Lec # 39)**

82. Which one of the formula will be used to find out the confidence interval for μ , when population variance unknown and sample size is large?

- o $\bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$ **PG # 268**
- o $\bar{x} \pm t_{\alpha/2(v)} \cdot \frac{s}{\sqrt{n}}$
- o $\bar{x} \pm t_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$
- o $\bar{x} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n-1}}$

83. In the test of goodness of fit, the _____is used as a test statistic.

- o F
- o T
- o Z
- o x^2 **PG # 332 (Lec # 44)**

84. If there are 8 treatments with 6 blocks in a randomized completed block design then what are the degrees of freedom for treatments?

- 5
- 4
- 6
- **7**

According to formula "a" is denote the Treatment of Degree of Freedom, a-1 means number of treatment minus 1, 8-1=7

Source	Degrees of Freedom
Treatment	a-1
Blocks	b-1
Residual	(a-1)(b-1)

85. Two continuous r.v."s X and Y are said to be independent if and only if :

- $f(x,y)=g(x)h(y)$ **PG # 200 (Lec # 26)**
- $f(x,y) \neq g(x)h(y)$
- $f(x,y)>g(x)h(y)$
- $f(x,y)<g(x)h(y)$

86. The lottery tickets issued for the purpose of money-making follows a:

- Normal distribution
- **Discrete uniform distribution PG # 210 (Lec # 27)**
- Binomial distribution
- Hypergeometric distribution

87. Uniform distribution is defined by:

- Largest value
- **Largest and smallest value**
- Smallest value
- Central value

88. Quantitative variable is further divided into:

- Continuous variable
- Discrete variable
- **Continuous & Discrete variable** **PG # 9 (Lec # 1)**
- None of the above

89. Color of the dress is the example of:

- **Qualitative data**
- Quantitative data
- Continuous data
- Discrete data

Note: Qualitative data is information about qualities; information that can't actually be measured. Some examples of qualitative data are the softness of your skin, the grace with which you run, and the color of your eyes.

90. Which one is commonly called a bell shaped distribution?

- **Symmetrical**
- Bimodal
- Skewed
- U shaped

Note: See Mcq 72

91. A fair coin is tossed three times, the probability that at least one head appear is:

- 1/2
- 1/8
- 6/8
- **7/8** **PG # 150 (Lec # 18)**

92. The probability of simultaneous occurrence of two events is called:

- Subjective probability
- Conditional probability
- **Joint probability** **PG # 194**
- Prior probability

Note: The simultaneous occurrence of two events A and B is called a JOINT event. The probability of a joint event is called a JOINT PROBABILITY.

93. What is the stem part of 243:

- 3
- 43
- 23
- **24** **PG # 47 (Lec # 6)**

94. A numerical value used as a summary measure for a sample, such as sample mean, is known as a:

- Population Parameter
- Sample Parameter
- **Sample Statistic** **Click Here For Reference**
- Population mean

95. Given the series 1,2,1,1,2,2,2,2,3,4,5,3,2,3,1,4,2,3.Which one of the following is mode of the given series:

- 4
- 4
  3
  2
- 3
  2
  1
- 3
  2*

Note: Mode means repeating number in a series.

96. P (A or B) =P (A) +P (B), then A and B are:

- **Mutually exclusive events** **PG # 154 (Lec #19)**
- Independent events
- Exhaustive events
- Equally likely events

97. First moment about origin is always equals to:

- Mean
- Variance
- Standard Deviation
- **Zero** **PG # 119**

98. E(4X + 5) = _____

- 16 E (X)
- 16 E (X) + 5
- 12 E (X)
- **4 E (X) + 5** **PG # 192 (Lec # 25)**

99. If σ^2 is unknown, then we use Z-test if the sample size is:

- o $n \geq 30$
- o $n < 30$
- o n=25
- o n=20

100. When a coin is tossed 3 times, the probability of 3 tails is:

- o **1/8**
- o 2/4
- o 3/8
- o 2/8

101. The F-distribution has……. Parameter.

- o One
- o No
- o **Two** **PG # 312 (Lec # 42)**
- o Three

F-distribution has two parameters $\nu 1$ and $\nu 2$, which are known as the degrees of freedom of the F-distribution.

102. Which one of the following provides the basis for hypothesis testing?

- o Null hypothesis
- o Alternative hypothesis
- o Critical value
- o **Test-statistic**

103. Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the alternative hypothesis is:

- o $H_0 : \mu = 100$
- o $H_0 : \mu > 100$
- o $H_0 : \mu < 100$
- o $H_1 : \mu \neq 100$ **PG # 278 (Lec # 36)**

104. By definition $f(x_i|y_j)$ = _____

- o $\frac{f(x_i, y_j)}{h(y_j)}$
- o $\frac{f(x_i, y_j)}{h(x_i)}$
- o $f(x_i, y_j)$
- o $f(y_j)$

Note: The conditional function $f(x_i|y_j)$ gives the probability of the values of x_i when $y = y_j$, $f(x_i|y_j)$ = $\frac{f(x_i, y_j)}{h(y_j)}$

105. Which one of the following is the most common example of a situation for which the main parameter of interest is a population proportion?

- o An observational study
- o A normal experiment
- o A randomized experiment
- o **A binomial experiment**

106. An estimator which has the smallest standard error among all unbiased estimators fulfills the property of_____

- o Unbiasedness
- o **Efficiency** **PG # 261 (Lec # 34)**
- o Consistency
- o Sufficiency

107. Which of the following can never be taken as the probability of an event?

- o 1
- o 0
- o 0.5
- o **-0.5**

Note: Probability always lie between 0 and 1

108. A set is any well-defined collection of

- Positive Objects
- Negative Objects
- Same Objects
- **Distinct Objects** **PG # 133 (Lec # 16)**

109. Measure of dispersion is used to calculate the:

- Central value
- Highest value
- Lowest value
- **Scattered value** **PG # 82**

110. If X and Y are independent variables then Var (X-Y) =

- **Var(X) – Var(Y)**
- Var(X) + Var(Y)
- Var(X+Y)
- Var(X) x Var(Y)

111. How the standard error is decreased :

- By decreasing the sample size
- By decreasing the mean
- By increasing the standard deviation
- **By increasing the sample size**

Note: The standard error decreases as the sample size increases.

112. For $\alpha = 0.01$, the critical values of z for two tailed test are equal to:

- -2.58 and+2.58
- -2.33 and+2.33
- -1.645 and +1.645
- **1.96 and + 1.96** **PG # 279 (Lec # 37)**

113. "A point estimate plus/minus a few times the standard error of that estimate". This statement represents:

- **Confidence interval** **PG # 270 (Lec # 35)**
- Critical region
- Acceptance region
- Critical value

114. The proportion of males in Pakistan is at least 0.48, the alternative hypothesis H_1 is

- $P \leq 0.48$
- $P = 0.48$
- $P < 0.48$ **PG # 278**
- $P \geq 0.48$

115. To find the confidence interval for the ratio of two variances, we use

- **F-Distribution**
- Z-Distribution
- Chi-square-Distribution
- t-Distribution

Note: Two variances:
F-distribution behind us, let's again jump right in by stating the confidence interval for the ratio of two population variances.

116. The Chi-Square distribution is continuous distribution ranging from:

- $-\infty \leq x^2 \leq \infty$
- $-\infty \leq x^2 \leq 1$
- $-\infty \leq x^2 \leq 0$
- $0 \leq x^2 \leq \infty$ **PG # 307 (Lec # 41)**

117. In a binomial experiment the total number of trials are:

- **Fixed in advance** **PG # 211 (Lec # 27)**
- Changeable according to situation
- Unpredictable
- Not independent

118. Which of the following value could not represent a coefficient of correlation?

- r = 0.99
- **r = 1.09** **PG # 128**
- r = -0.73
- r = -1

Note: r is a pure number that lies between and i.e. **-1< r < 1**

119. In a one way ANOVA test there are 5 observations in each of three treatments. The degrees of freedom for the treatments is:

- o 5
- o 3
- o 1
- o **2**

120. If P(B|A) = 0.25 and $P(A \cap B) = 0.20$, then P(A) =

- o 0.05
- o **0.80**
- o 0.95
- o 0.75

Note: $P(B|A) = \frac{P(A \cap B)}{P(A)}$ in that case we $\frac{P(A \cap B)}{P(B \mid A)} = \frac{0.20}{}$ $= 0.8$

121. If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- o **0,1,2,3**
- o 1,3,3,1
- o 1, 2, 3
- o 3, 2

122. When f(x) is continuous probability function, then P(X = 2) is:

- o 1
- o 0.5
- o **0**
- o 0.25

123. Atmosphere pressure is the example of:

- o Constant
- o Qualitative variable
- o **Quantitative variable** **PG # 8 (Lec # 1)**
- o None of the above

124. Which of the following scale has true zero point?

- **Ratio Scale** **PG # 9 (Lec # 1)**
- Interval scale
- Nominal scale
- Ordinal scale

125. Given the series 1,2,1,1,2,,2,2,2,3,3,4,5,3,2,3,1,4,2,3. Which one of the following is mode of the given seires:

- 4
- 3
  3
  2
- 3
  2
- 1
  3
  2*

Note: Mode means repeating number in a series.

126. Var (4X+ 5) = _____

- 16 Var (X) may be
- 16 Var (Xj+ 5
- **4 Var (X) + 5**
- 12 Var (X)

127. The hyper geometric random variable is a(an):

- Continuous variable
- **Discrete variable**
- Undefined
- Independent variable

Note: The probability mass function of the discrete random variable *X* is called the **hypergeometric distribution**

128. Assuming that following is a probability distribution, then what is the value of „a‟ :

X	1	2	3
P(X)	0.1	a	0.1

- o 0.6
- o **0.8**
- o 0.2
- o 0.4

129. If f (x, y) is bivariate probability density function of continuous random variables X and Y then marginal density function of y i.e *h(y)*is:

- o $\int_{-\infty}^{\infty} f(x,y)dx$ **PG # 199**
- o $\int_{-\infty}^{\infty} f(x,y)dy$
- o $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)dx\ dy$
- o $\int_{a}^{b}\int_{c}^{d} f(x,y)dy\ dx$

130. _____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

- o Hypothesis
- o Lower limit
- o **Confidence interval**
- o Probability limit

131. Test Statistics x^2 is equal to:

- o $\frac{n^2S^2}{\sigma_0^2}$
- o $\frac{nS^2}{\sigma_0^2}$ **PG # 309**
- o $\frac{S^2}{2\sigma_0^2}$
- o $\frac{n^2S^2}{2\sigma_0^2}$

132. The LSD test is applied when the null hypothesis is:

- **Rejected** **PG # 331 (Lec # 44)**
- Accepted
- Finalized
- Acknowledged

133. What is the probability of drawing a red-queen card from a well shuffled pack of 52 playing cards?

- 4/52
- **2/52**
- 13/52
- 26/52

Note: In a pack of 52 playing cards there are 2 red-queen cards

134. The probability of drawing a spade card is:

- 1/52
- 4/52
- **13/52**
- 26/52

Note: Total spade cards in pack of 52 playing card is 13.

135. In construction of a histogram, what would be taken along X-axis?

- Midpoints
- Class limits
- Class interval
- **Class boundaries**

Note: A histogram consists of a set of adjacent rectangles whose bases are marked off by class boundaries along the X-axis, and whose heights are proportional to the frequencies associated with the respective classes.

136. If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- Ogive
- **Frequency polygon** **Click Here For Reference**
- Frequency curve
- Historigram

137. Which type of the curve is represented by the following shape?



- **Negatively skewed curve** **PG # 39 (Lec # 5)**
- Bell shape curve
- semi-symmetrical curve
- Positively skewed curve

138. If A = (1 , 2, 3, 4, 5 , 10) and B = {1, 3, 5} then $B \subset A$ means:

- A is less than B
- A is contained in B
- **B is contained in A.** **PG # 134 (Lec # 16)**
- B is less than A

139. What is the mode in the word STATISTICS:

- S
- T
- T
- **S and T**

Note: Most Repeated value in a data set.

140. In a symmetrical distribution, if $Q_1 = 20,\ Q_2 = 30$ *then* Q_3 *is* :

- **50**
- 40
- 30
- 10

$Q_1 + Q_2 = Q_3 + 20 + 30 = 50$

141. Which one of the following is not a included in measures of central tendency:

- **Quartile Deviation** **PG # 51 (Lec # 6)**
- Harmonic Mean
- Geometric Mean
- Arithmetic Mean

Note: Quartile Deviation is not included in measures of central tendency because in measure of central tendency (i.e. an average) indicates the location or general position of the distribution on the X axis, it is also known as a measure of location or position. Whereas **Quartile Deviation** is defined as half of the difference between the third and first quartiles.

142. The sum of deviations from mean is:

- Maximum
- Minimum
- **Zero** **PG # 86 (Lec # 10)**
- Undefined

143. Mean deviation is always:

- Less than Standard Deviation
- Greater than Standard Deviation
- Greater or equal to Standard Deviation
- **Less or equal to Standard Deviation**

144. The probability of an event always lies between:

- 0 and $-\infty$
- -1 and +1
- $-\infty$ and $+\infty$
- **0 and 1**

Note: The probability of any event lies between **0** and 1

145. The number of parameters in a Poisson distribution is (are):

- **1** **PG # 222**
- 0
- 2
- 3

The Poisson distribution has only one parameter $\mu > 0$. The parameter μ may be interpreted as the mean of the distribution.

146. The degrees of freedom for a t-test with sample size 14 is:

- o 14
- o **13**
- o 7
- o 0

Note: t-distribution with v = n – 1 degrees of freedom

147. The degrees of freedom for a t-test with sample size 6 is:

- o 1
- o 3
- o **5**
- o 7

Note: t-distribution with v = n – 1 degrees of freedom

148. In normal distribution: β_1=……..:

- o 0 **PG #277 (Lec # 30)**
- o 1
- o 2
- o 3

149. Which of the following is a measure of absolute dispersion?

- o Skewness
- o Mean Deviation
- o **Coefficient of variation** **PG #92 (Lec # 11)**
- o Kurtosis

150. An automobile is running, during the first 60 Km, at the rate of 10 Km/hr. During the second 60 Km at the rate of 30Km/hr, while during the third 60 Km its speed was 40 Km/hr. What method is more appropriate to calculate the average speed?

- o Median
- o Arithematic mean
- o **Harmonic mean** **PG # 78 (Lec # 9)**
- o Geometric mean

151. If S.D (X) = 5 then S.D $(\frac{2X+5}{2})$ = _____

- 5
- 10
- 15
- **7.5**

Note: 2 multiply by the value of X which is $(\frac{2X+5}{2})$ according to the calculation ans is 7.5

152. For a particular data set the Pearson"s coefficient of skewness is greater then zero. What will be the shape of distribution?

- Negatively skewed
- J-shaped
- Symmetrical
- **Positively skewed** **PG # 111 (Lec # 14)**

153. Which of the following is correct option to find $P(X + Y \leq 1)$?

- f(0 , 0) + f(0,1) + f(1,2)
- f(2,0) + f(0, 1) + f(1, 0)
- f(0, 0) +f (1, 1) + f(1, 0)
- **f (0,0)+ f (0,1) + f (1,0)** **PG # 197 (Lec # 26)**

154. If a significance level of 1% is used rather than 5%, the null hypothesis is:

- More likely to be rejected
- **Less likely to be rejected** **Click Here For Reference**
- Just as likely to be rejected
- None of the above

155. The covariance of a random variable with itself is:

- Zero
- One
- **Its variance**
- Its correlation

156. For any two estimators T1 and T2, if VAR(T1) < VAR(T2), then T1 is:

- Consistent
- Unbaised
- Sufficient
- **Efficient** **PG #261 (Lec # 34)**

157. Which type of the curve is represented by the following shape?



- Bell shape curve
- semi-symmetrical curve
- Positively skewed curve
- **Negatively skewed curve** **PG # 39 (Lec # 5)**

158. Which one of the following is not a type of frequency curve?

- The symmetrical frequency curve
- The extremely skewed frequency curve
- The U-shaped frequency curve
- **Frequency polygon** **PG # 38 (Lec # 5)**

159. For positively skewed distribution

Mean…… Median…….. Mode:

- ≠
- =
- <
- >

Note: Mean is less than the median, and they are both less than the mode. The mean and the median both reflect the skewing, but the mean reflects it more so.

160. What is „ f_m “ in the formula of mode:

- First Frequency
- Last Frequency
- **Maximum Frequency**
- Minimum Frequency

161. In a set of 10 values all the values are 5, what will be the P_{50}?

- o 2
- o **5** **Click Here For Reference**
- o 10
- o 20

162. Which of the following is true for the Poisson distribution:

- o mean > variance
- o mean < variance
- o **mean = variance** **PG # 223 (Lec # 29)**
- o mean = standard deviation

163. Under what condition would you use the paired t-test?

- o When there is a single sample of data
- o When the two samples of data arc independent
- o **When A the two samples of data are not independent** not sure
- o When there are two proportions

Note: 2-sample *t*-test treating them as independent samples, you will not be able to reject the null hypothesis. This demonstrates the importance of distinguishing the two types of samples. Also, it is wise to design an experiment efficiently whenever possible.

164. In the case of the sampling distribution of $\bar{X}$, the finite population correction factor (fpc) is:

- o $\sqrt{\frac{n-N}{N-1}}$
- o $\sqrt{\frac{N-1}{N-n}}$
- o $\sqrt{\frac{N-n}{N-1}}$ **PG # 248 (Lec # 32)**
- o $\sqrt{\frac{N-n}{N}}$

165. The collection of all the elements under study is called:

- o **Population** **PG # 12 (Lec # 2)**
- o Sample
- o Data
- o Registration

166. If E is an impossible event, then P(E)is:

- 1
- 0.15
- **0** **Click Here For Reference**
- 0.5

167. For the given set of observations 1, 4, 4, 4 and 7, which statement is true:

- *mean > median > mode*
- ***mean= median = mode***
- *mean ≠ median ≠ mode*
- *mean< median < mode*

168. Which one is the formula for calculating the variance of the t-distribution?

- $\sigma^2 = \sqrt{\frac{v}{v-2}}$
- $\sigma^2 = \frac{v^2}{v-2}$
- $\sigma^2 = \frac{v^2}{v-1}$
- $\sigma^2 = \frac{v}{v-2}$ **PG # 293 (Lec # 39)**

169. If a coin is tossed 3 times then what are the total number of sample points in the sample space?

- 3
- 8
- 6
- 4

Note: $2^3 = 8$, 2x2x2 = 8

170. Which one of the following is the class frequency?

- **The number of observations in each class**
- The difference between consecutive lower class limits
- Always contains at least 5 observations
- Usually a multiple of the lower limit of the first class

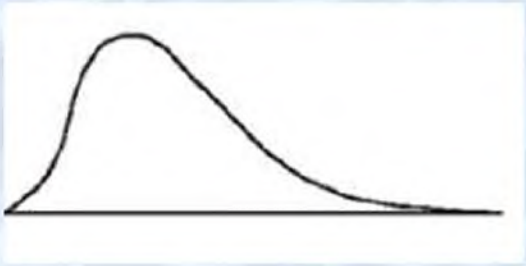
171. Which of the following scale has not true zero point?

- Nominal scale
- Ordinal Scale
- **Interval scale** **PG # 9 (Lec # 1)**
- Ratio scale

172. Data obtained from the Bureau of Statistics, is an example of:

- Primary data
- **Secondary data** **PG # 12 (Lec # 2)**
- Qualitative data
- None of these

173. What is the name of following curve?



- Symmetrical curve
- Normal curve
- Negatively skewed curve
- **Positively skewed curve** **PG # 39 (Lec # 5)**

174. The end points of a confidence interval are called:

- Confidence coefficient
- Confidence interval
- **Confidence limits**
- Parameters

175. Let X and Y are two random variables. Which property of expectation is true?

- **E(X – 3Y) = 2E(X) - 3E(Y)**
- E(2X - 3Y) = 2E(X) - 3E(Y) +E(XY)
- E(2X - 3Y) = 2E(X) + 3E(Y) -E(XY)
- E(2X -3Y) = 2E(X) - 3E(Y) - 2 E(XY)

176. The main objective of non-central distribution is:

- **To minimize variance**
- Testing of hypothesis
- To increase variance
- None of these

177. A statement about the population parameter (unknown), which may or may not be true is called

- Test statistics
- **Statistical hypothesis** **PG # 277 (Lec # 36)**
- Null hypothesis
- Alternative hypothesis

178. A Bernoulli trial has:

- At least two outcomes
- At most two outcomes
- **Two outcomes**
- Fewer than two outcomes

Note: a Bernoulli trial (or binomial trial) is a random experiment with exactly two possible outcomes, "success" and "failure".

179. If b(x, 5, 0.398), then the variance of this distribution is:

- **1.198** **Click Here For More Detail**
- 3.51
- 1.012
- 3.21

180. If the population standard deviation a is doubles, the width of the confidence interval for the population mean μ (i.e. ; the upper limit of the confidence interval-lower limit of the confidence interval) will be:

- Divided by 2
- Multiplied by
- **Double**
- Decrease

because the length of the interval is proportional to

181. If n1 = 16, n2 = 9 and a = 0.01; t_{α} equals:

- 2.737
- **2.807**
- **2.797**
- 3.767

182. If the population standard deviation σ is unknown and the sample size n is less than or equal to 30, the confidence interval for the population mean μ is:

- $p \pm Z_{\alpha 2}\sqrt{\frac{\hat{pq}}{n}}$
- $\overline{X} \pm Z_{\alpha 2}\frac{s}{\sqrt{n}}$
- $\overline{X} \pm t_{\alpha 2^{(v)}}\frac{s}{\sqrt{n}}$ **PG # 297**
- $\overline{X} \pm t_{\alpha 2^{(v)}}\frac{S_d}{\sqrt{n}}$

183. The number of values that are free to vary after we have placed certain restrictions upon the data is called:

- **Degrees of freedom**
- Confidence coefficient
  Confidence coefficient
- Number of parameters
- Number of samples

Note: Degree of freedom are the number of values that are free to vary after certain restrictions have been imposed on all values

184. In a hyper geometric distribution N=6 ,n=2,K=3 then means is

- 2
- 3
- **1**
- 4

Note: The mean of the distribution is equal to $n \times k / N$ which is 2×3/6=1

185. In a Normal distribution $N(\mu,\sigma^2)$, μ_4 is equal to

- o σ_4
- o $3\sigma^4$ **PG # 228**
- o $\frac{3\sigma^4}{4}$
- o $\frac{\sigma^4}{2}$

Note: In Normal distribution $\mu_4 = 3\sigma_4$

186. Suppose that a population has mean $\mu = 24$ what is the mean of the sampling A distribution of the mean for samples of size n=25?

- o 24
- o 2
- o **4.8**
- o 25

Note: According to formula $\frac{\mu}{\sqrt{n}}$ **=4.8**

187. In sampling with replacement the standard error of sample proportion p is equal to:

- o $\frac{p(1-p)}{n}$ **Click Here For Reference**
- o $\sqrt{\frac{p+q}{2}}$
- o $\sqrt{\frac{p(1-p}{n}\frac{N-n}{N-1}}$
- o $\sqrt{\frac{p(1-p}{n}}$

188. From point estimation, we always get:

- o **Single value** **PG # 257 (Lec # 33)**
- o Two values
- o Range of values
- o Zero

189. F-distribution is…….

- **Positively skewed** **PG # 312 (Lec # 42)**
- Negatively skewed
- Normal
- Symmetrical

190. If the outcome is an odd number when a die is rolled then what is the probability of *prime number :*

- **2/3**
- 1/3
- 1/6
- 5/6

Note: Sample space, S = {1,2,3,4,5,6}, Let A be the event of getting an odd number, Let B be the event of getting a prime number. A = {1,3,5}, B = {2,3,5}, A ∩ B = {3, 5} conditional probability of event B given that event A has already occurred, P(B|A) = P(A∩B)/P(A)=(2/6)/(3/6)=2/3

191. The sample variance $S^2 = \frac{\sum (x - \bar{x})^2}{n}$ is:

- **Unbiased estimator of σ^2** **PG # 264 (Lec # 34)**
- Biased estimator of σ^2
- Unbiased estimator of μ
- Biased estimator of μ

192. _____ are the values that make the boundaries of the confidence interval.

- Confidence intervals
- **Confidence limits**
- Levels of confidence
- Margin of error

193. If $\sum_{i=1}^{5} (X_i - 20) = 0, then \overline{X}$ …….

- **0**
- 20
- 5
- 25

194. The conditional probability is P(A|B) is given by formula:

- o $\frac{P(A \cup B)}{P(A)}$
- o $\frac{P(A \cap B)}{P(B)}$ **PG # 159 (Lec # 20)**
- o $\frac{P(A \cap B)}{P(A)}$
- o $\frac{P(A \cup B)}{P(B)}$

195. In statistics LSD is the abbreviation of…….

- o Large sample difference
- o **Least significant difference** **PG # 328 (Lec # 44)**
- o Least significant digit
- o Low speed data

196. If the random variable X denotes the number of heads when three distinct coins are tossed, then X assumes the value

- o **0, 1, 2, 3** **Click Here For Reference** **PG # 167**
- o 1, 3,3, 1
- o 1, 2, 3
- o 3 only

197. In a set of 20 values all the values are 5, what will be the mean?

- o 4
- o **5**
- o 10
- o 20

198. E(4X + 5) = _____

- o 12 E (X)
- o **4 E (X) + 5** **PG # 192 (Lec # 25)**
- o 16 E (X) + 5
- o 16 E (X)

199. Which one of the following assumptions is not a requirement for ANOVA?

- **Dependent samples**
- Normal populations
- random samples
- Independent samples

Note: assumptions for ANOVA

Each group sample is drawn from a normally distributed population

- **All populations have a common variance**
- **All samples are drawn independently of each other**
- **Within each sample, the observations are sampled randomly and independently of each other**
- **Factor effects are additive**

200. Mean is affected by the change of:

- Origin only
- Scale only
- **Origin and Scale**
- Not affected

Note: For calculating the mean, changing the origin for your data means it will affect the mean.

201. For two independent events A and B if P (A) = 0.2 and P(B) = 0.4 ,then what is $P(A \cap B)$?

- 0.06
- **0.08**
- 0.02
- 0.04

Note: $P(A \cap B) = P(A) \times P(B) = 0.2 \times 0.4 = 0.08$

202. The number of parameters in hypergeometric distribution is (are):

- 1
- 2
- **3** **PG # 219 (Lec # 28)**
- 4

203. In a symmetric distribution, what percent of values are less than the median?

- o 30%
- o **50%**
- o 70%
- o 90%

Note: Symmetric distribution also called normal distribution The mean, median, and mode of a normal distribution are equal.

204. If the probability of one event does not affected by the occurrence of another event then both events are:

- o Dependent
- o Not Mutually Exclusive
- o Mutually Exclusive
- o **Independent**

Note: Independent Events
Two events are independent if the occurrence of one does not affect the probability of the other occurring.

205. For the Poisson distribution $P(X=1)=\frac{e^{-2}2^{1}}{1!}$ the mean value is:

- o 1
- o **2**
- o 3
- o 0

206. The degrees of freedom for a t-test with sample size 10 is:

- o 5
- o 8
- o 9
- o **10**

Note: The sum of all values in the data must equal *n* x mean, where *n* is the number of values in the data set.

207. In normal distribution M.D. =

- o 0.5 σ
- o 0.75σ
- o **0.7979σ**
- o 0.6445σ

208. _____is the standard deviation of a sampling distribution.

- **Standard error**
- Sample standard deviation
- Replication error
- Sampling error

Note: Suppose that we draw all possible samples of size *n* from a given population. Suppose further that we compute a statistics e.g., a mean, proportion, standard deviation) for each sample. The probability distribution of this statistic is called a sampling distribution. And the standard deviation of this statistic is called the standard error.

209. For what condition of k, at least $1-1/k^2$ of the data-values fall within k standard deviations of the mean:

- **Greater than 1** **PG # 94 (Lec # 12)**
- Less than 1
- Greater or equal to 1
- Less or equal to 1

210. If a player well shuffles the pack of 52 playing card then the probability of a black card from 52 playing cards is:

- 1/52
- 13/52
- 4/52
- **26/52**

Note: Total pack of playing card is 52, 26 red cards and 26 black, or 52/2=26

211. Interval estimation and confidence interval are:

- Same
- **Different** **(PG # 264 Lec # 34)**
- Opposite
- Independent

Note: We will have no way of ascertaining how close this particular is to μ .Whereas a point estimate is a single value that acts as an estimate of the population parameter, interval estimation is a procedure of estimation the unknown parameter which specifies a range of values within which the parameter is expected to lie. A confidence interval is an interval computed from the sample observation x ,x2……xn, with a statement of how confident we are that interval does contain the population parameter.

212. The average height of Pakistani solders exceeds the average height of American solders by not more than 2 inches. The alternative hypothesis H_1 is

- $\mu_P - \mu_A = 2^n$
- $\mu_P - \mu_A \geq 2^n$
- $\mu_P - \mu_A > 2^n$
- $\mu_P - \mu_A < 2^n$ **(PG # 278 Lec # 36)**

213. The proportion of working females in Pakistan is at most 0.30, the alternative hypothesis.

- $P < 0.30$ **(PG # 278 Lec # 36)**
- $P = 0.30$
- $P \leq 0.30$
- $P > 0.30$

214. For degree of freedom $v \leq 2$ the variance of t-distribution:

- Is greater than zero
- Is less than one
- Is equal to one
- **Does not exist** **PG # 294 (Lec # 39)**

215. All possible trials by throwing a dice is an example of:

- Finite population
- Infinite population
- **Hypothetical population** **PG # 12 & 13**
- Limited population

216. The sum of squares of deviations from mean is:

- Undefined
- Maximum
- **Minimum** **Click Here For Detail**
- Zero

217. The critical region for $H_1 : \mu < \mu_0$ *when* $\alpha = 0.01$ is

- $z > z_{0.01}$ **PG # 281 (Lec # 37)**
- $|z| > z_{0.01}$
- $z < -z_{0.05}$
- $|z| > z_{0.05}$

Note: Give me a feedback and your Suggestion also If you find any mistake in mcqz plz inform me Via Contact us Page on our Site. And tell me your answer with references.

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FINAL TERM EXAMINATION
Spring 2012
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
For a particular data the value of Pearson's coefficient of skewness is greater then zero. What will be the shape of distribution?

► Negatively skewed
► J-shaped
► Symmetrical
► **Positively skewed (Page 109)**

Question No: 2 (Marks: 1) - Please choose one
In measures of relative dispersion unit of measurement is:

► Changed
► **Vanish**
► Does not changed
► Dependent

Question No: 3 (Marks: 1) - Please choose one
The F-distribution always ranges from:

► 0 to 1
► 0 to -∞
► -∞ to +∞
► **0 to +∞ (Page 312)**

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے

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Question No: 4 (Marks: 1) - Please choose one
In chi-square test of independence the degrees of freedom are:
► **n - p**
► n - p-1
► n - p- 2
► n – 2

Question No: 5 (Marks: 1) - Please choose one
The Chi- Square distribution is continuous distribution ranging from:
► $-\infty \leq \chi 2 \leq \infty$
► $-\infty \leq \chi 2 \leq 1$
► $-\infty \leq \chi 2 \leq 0$
► **$0 \leq \chi 2 \leq \infty$ (Page 307)**

Question No: 6 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

► $E(X)+E(Y)$
► $E(X)-E(Y)$ **(Page 202)**
► $X-E(Y)$
► $E(X)-Y$

Question No: 7 (Marks: 1) - Please choose one
If ŷ is the predicted value for a given x-value and b is the y-intercept then the equation of a regression line for an independent variable x and a dependent variable y is:

► **ŷ = mx + b, where m = slope (Page 121)**
► x = ŷ + mb, where m = slope
► ŷ = x/m + b, where m = slope
► ŷ = x + mb, where m = slope

Question No: 8 (Marks: 1) - Please choose one
The location of the critical region depends upon:

► Null hypothesis
► **Alternative hypothesis (Page 281)**
► Value of alpha
► Value of test-statistic

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FINAL TERM EXAMINATION
Spring 2012
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
The t-Distribution is**…….** Spread out then the standard normal Distribution.
► Less
►**More (Page 293)**
►Equally
►Not

Question No: 2 (Marks: 1) - Please choose one
To find the confidence internal for the ratio of two variances we use:
►**F-Distribution (Page 311)**
►Z-Distribution
►Chi-Square Distribution
►T-Distribution

Question No: 3 (Marks: 1) - Please choose one
How many percent of values are less then 4th deciles in a symmetric distribution.
►14
►24
►4
►**40 (Page 70)**

Question No: 4 (Marks: 1) - Please choose one
The combined distribution of more than two random variables is:
►Bivariate Distribution
►Marginal Distribution
►**Joint Distribution (Page 194)**
►Univariate Distribution

Question No: 5 (Marks: 1) - Please choose one
The degrees of freedom for a T-test with sample size 14 is:
►14
►**13 (Page 341)**
►**7**
►0

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

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Question No: 6 (Marks: 1) - Please choose one
Which of the falling is true for the binomial distribution b(x: n, p):
►**Mean > Variance (Page 214)**
►Mean < Variance
►Mean = Variance
►Mean = Standard Deviation

Question No: 7 (Marks: 1) - Please choose one
What is *m f* in the formula of mode?
►first frequency
►last frequency
►middle frequency
►**highest frequency (Page 54)**

FINAL TERM EXAMINATION
Fall 2011
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
The parameters of the binomial distribution b(x; n, p) are:

► x & n
► x & p
► **n & p (Page 212)**
► x, n & p

Question No: 2 (Marks: 1) - Please choose one
Which of the following is true for the Poisson distribution:

► mean > variance
► mean < variance
► **mean = variance (Page 223)**
►mean= standard deviation



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Question No: 3 (Marks: 1) - Please choose one
If a significance level of 1% is used rather than 5%, the null hypothesis is:

- ► More likely to be rejected
- ► **Less likely to be rejected** Click here for detail
- ► Just as likely to be rejected
- ► None of the above

Question No: 4 (Marks: 1) - Please choose one
The variance of the chi-square distribution is:

- ► $2v$ **(Page 307)**
- ► $v-1$
- ► $v-2$
- ► v

Question No: 5 (Marks: 1) - Please choose one
The degrees of freedom for a t-test with sample size 10 is:

- ►5
- ► 8
- ► **9 (Page 298)**
- ► 10

FINAL TERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 4)

Question No: 1 (Marks: 1) - Please choose one
The value of χ^2 can never be :

- ► Zero–
- ► Less than 1
- ► Greater than 1
- ► **Negative (Page 307)**

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

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Question No: 2 (Marks: 1) - Please choose one
The mean of the F-distribution is:

- $\frac{v_1}{v_1 - 2} \quad for v_1 \rangle 2$
- $\frac{v_2}{v_2 - 2} \quad for v_2 \rangle 2$ **(Page 312)**
- $\frac{v_1}{v_1 - 2} \quad for v_1 \geq 2$
- $\frac{v_2}{v_2 - 2} \quad for v_1 \leq 2$
-

Question No: 3 (Marks: 1) - Please choose one
The F-distribution always ranges from:

- 0 to 1
- 0 to -∞
- -∞ to +∞
- **0 to +∞ (Page 312) rep**

Question No: 4 (Marks: 1) - Please choose one
The total number of samples when sampling is done with replacement :

- N^n **(Page 237)**
- C_n^N
- $\frac{N-n}{N-1}$
-
- 1

زندگی میں کامیابی کا یہی راز ہے کہ پریشانیوں سے پریشان مت بنو

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Question No: 5 (Marks: 1) - Please choose one
ANOVA was introduced by :

- Helmert
- Pearson
- **R.A Fisher (Page 320)**
- Francis

Question No: 6 (Marks: 1) - Please choose one
The test statistic used in analysis of variance procedure follow the ..... distribution.:

- χ^2
- T
- Z
- **F (Page 326)**

Question No: 7 (Marks: 1) - Please choose one
For testing of hypothesis about population proportion , we use:

- **Z-test (Page 292)**
- t-Test
- Both Z & T-test
- F test

Question No: 8 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

- $E(X)+E(Y)$
- $E(X)-E(Y)$ **(Page 202) rep**
- $X-E(Y)$
- $E(X)-Y$

Question No: 9 (Marks: 1) - Please choose one
A die is rolled. What is the probability that the number rolled is greater than 2 and even:

- 1/2
- **1/3 2/6 = 1/3**
- 2/3
- 5/6

Hint: - number that is greater than 2 and is even = 4,6 i.e only 2 numbers out of 6
Therefore the probability 2/6 = 1/3

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Question No: 10 (Marks: 1) - Please choose one

The probability of drawing a king of spade from a pack of 52 cards is:

- ► 1/4
- ► 1/13
- ► 1/26
- ► **1/52**

Question No: 11 (Marks: 1) - Please choose one

An estimator T is said to be unbiased estimator of θ if

- ► **E (T) = θ (Page 258)**
- ► E (T) =T
- ► E (T) =0
- ► E (T) =1

Question No: 12 (Marks: 1) - Please choose one

From point estimation, we always get:

- ► **Single value (Page 257)**
- ► Two values
- ► Range of values
- ► Zero

Question No: 13 (Marks: 1) - Please choose one

The best unbiased estimator for population variance σ^2 is:

- ► Sample mean
- ► Sample median
- ► Sample proportion
- ► **Sample variance (Page 260)**

Question No: 14 (Marks: 1) - Please choose one

When c is a constant, then E(c) is:

- ► 0
- ► 1
- ► **c (Page 180)**
- ► -c



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Question No: 15 (Marks: 1) - Please choose one
Var(4X + 5) =_____

- 16 Var (X)
- **16 Var (X) + 5 (correct)**
- 4 Var (X) + 5
- 12 Var (X)

Question No: 16 (Marks: 1) - Please choose one
When f(x) is continuous probability function, then P(X = 1) is:

- 1
- ∞
- −∞
- **0 (Page 188)**

Question No: 17 (Marks: 1) - Please choose one
The hyper geometric random variable is a(an):

- Continuous variable
- **Discrete variable** Click here for detail
- Undefined
- Independent variable

Question No: 18 (Marks: 1) - Please choose one
From a sample of 200 people were asked whether they like a particular product. Fifty said 'yes' and remain said 'no', assuming ‘yes’ means a success, which of the following is correct?

- Sample proportion p=0.33
- **Sample proportion p=0.25 (Page 245)**
- Population proportion p= 0.33
- Population proportion p=0.25

Question No: 19 (Marks: 1) - Please choose one
In any data set, what percent of values fall in the interval $Median \pm Q.D$?

- **50 per cent (Page 84)**
- 68.5 per cent
- 95.4 per cent
- 99 per cent

جھوٹ انسان اور ایمان دونوں کا دشمن ہے

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Question No: 20 (Marks: 1) - Please choose one

$$\sum_{i=1}^{5}(X_i - 20) = 0, then\ \bar{X} = ...$$

If

- **0 (Page 258)**
- 20
- 5
- 25

Question No: 21 (Marks: 1) - Please choose one

The height of a student is 60 inches. This is an example of .....?

- **Continuous data (correct)**
- Qualitative data
- Categorical data
- Discrete data

Question No: 22 (Marks: 1) - Please choose one

In Statistics, we have MSE which is abbreviation of……

- **Mean square error (Page 330)**
- Measured square error
- Medical screening exam
- Major sampling error

Question No: 23 (Marks: 1) - Please choose one

Which one is the formula of mid range:

- $x_m - x_0$
- $x_0 - x_m$
- $\frac{x_0 - x_m}{2}$
- $\frac{x_0 + x_m}{2}$ **(Page 80)**

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

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Question No: 24 (Marks: 1) - Please choose one
The deviation of a distribution from symmetry is called:

► Kurtosis
► **Skewness**
► Dispersion
► Flatness

Question No: 25 (Marks: 1) - Please choose one
If E is an impossible event, then P(E) is:

► 1
► 2
► **0 (Page 146)**
► 0.5

Question No: 26 (Marks: 1) - Please choose one
If a data set has the even number of observations, the median :
► Is the average value of the two middle items
► Can not be determined
► must be equal to the mean
► **Is the average value of the two middle items when all items are arranged in ascending order**
Click here for detail

Question No: 27 (Marks: 1) - Please choose one
For the Poisson distribution $P(X=1)=\frac{e^{-0.135}0.135^{1}}{1!}$ the mean value is :
► 2
► 5
►10
► **0.135 (Page 222)**

Question No: 28 (Marks: 1) - Please choose one
In testing of hypothesis, we always begin it with assuming that:

► **Null hypothesis is true (Page 277)**
► Alternative hypothesis is true
► Sample size is large
► Population is normal

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

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Question No: 29 (Marks: 1) - Please choose one

Variance of the t-distribution is given by the formula:

$$\sigma^2 = \sqrt{\frac{\nu}{\nu - 2}}$$

►

$$\sigma^2 = \frac{\nu^2}{\nu - 2}$$

►

$$\sigma^2 = \frac{\nu}{\nu - 1}$$

►

$$\sigma^2 = \frac{\nu}{\nu - 2}$$ **(Page 293)**

►

Question No: 30 (Marks: 1) - Please choose one

If a continuous probability distribution has $\beta_2 = 2.14$ then what will be peakedness of the distribution?

► **Platykurtic (Page 119)**
► Mesokurtic
► Leptokutic
► Moderately skewed

FINAL TERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 4)

Question No: 1 (Marks: 1) - Please choose one

When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

► Mutually exclusive
► **Equally likely (Page 117)**
► Not mutually exclusive
► Exhaustive

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

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Question No: 2 (Marks: 1) - Please choose one
The mean of the F-distribution is:

- $\frac{v_1}{v_1 - 2} \quad for v_1 \rangle 2$
- $\frac{v_2}{v_2 - 2} \quad for v_2 \rangle 2$ **(Page 312)**
- $\frac{v_1}{v_1 - 2} \quad for v_1 \geq 2$
- $\frac{v_2}{v_2 - 2} \quad for v_1 \leq 2$

Question No: 3 (Marks: 1) - Please choose one
The LSD test is applied only if the null hypothesis is:

- **Rejected (Page 331)**
- Accepted
- No conclusion
- Acknowledged

Question No: 4 (Marks: 1) - Please choose one
Analysis of variance is a procedure that enables us to test the equality of several:

- Variances
- **Means (Page 320)**
- Proportions
- Groups

Question No: 5 (Marks: 1) - Please choose one
ANOVA was introduced by :

- Helmert
- Pearson
- **R.A Fisher (Page 320) rep**
- Francis

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Question No: 6 (Marks: 1) - Please choose one
For testing of hypothesis about population proportion , we use:

- **Z-test (Page 292) rep**
- t-Test
- Both Z & T-test
- F test

Question No: 7 (Marks: 1) - Please choose one
If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- **0,1,2,3**
- 1,3,3,1
- 1, 2, 3
- 3, 2

Question No: 8 (Marks: 1) - Please choose one
If X and Y are independent variables, then E (XY) is:

- E(XX)
- **E(X).E(Y) (Page 202)**
- X.E(Y)
- Y.E(X)

Question No: 9 (Marks: 1) - Please choose one
The parameters of the binomial distribution b(x; n, p) are:

- x & n
- x & p
- **n & p (Page 212) rep**
- x, n & p

Question No: 10 (Marks: 1) - Please choose one
If P (E) is the probability that an event will occur, which of the following must be false:

- **P(E)= - 1**
- P(E)=1
- P(E)=1/2
- P(E)=1/3

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

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Question No: 11 (Marks: 1) - Please choose one

An estimator T is said to be unbiased estimator of θ if

- **► E (T) = θ (Page 258) rep**
- ► E (T) =T
- ► E (T) =0
- ► E (T) =1

Question No: 12 (Marks: 1) - Please choose one

The best unbiased estimator for population variance σ^2 is:

- **► Sample mean (Page 260) rep**
- ► Sample median
- ► Sample proportion
- ► Sample variance

Question No: 13 (Marks: 1) - Please choose one

$$S^2 = \frac{\sum(x-\bar{x})^2}{n}$$

The sample variance is:

- ► Unbiased estimator of σ^2
- **► Biased estimator of σ^2 (Page 260)**
- ► Unbiased estimator of μ
- ► None of these

Question No: 14 (Marks: 1) - Please choose one

When c is a constant, then E(c) is:

- ► 0
- ► 1
- **► c (Page 180) rep**
- ► -c

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

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Question No: 15 (Marks: 1) - Please choose one
If f (x, y) is bivariate probability density function of continuous r.v.'s X and Y then

$g(x)$ is:

- $\int_{-\infty}^{\infty} f(x,y)\,dx$
- $\int_{-\infty}^{\infty} f(x,y)\,dy$
- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$ **(Page 199)**
- $\int_{a}^{b}\int_{c}^{d} f(x,y)\,dy\,dx$
-

Question No: 16 (Marks: 1) - Please choose one
The analysis of variance technique is a method for :

- Comparing F distributions
- **Comparing three or more means (Page 320)**
- Measuring sampling error
- Comparing variances

Question No: 17 (Marks: 1) - Please choose one
The continuity correction factor is used when:

- The sample size is at least 5
- Both *nP* and *n (1-P)* are at least 30
- **A continuous distribution is used to approximate a discrete distribution** Click here for detail
- The standard normal distribution is applied

Question No: 18 (Marks: 1) - Please choose one
Stem and leaf is more informative when data is :

- Equal to 100
- Greater Than 100
- **Less than 100** click here for detail
- In all situations

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Question No: 19 (Marks: 1) - Please choose one
The branch of Statistics that is concerned with the procedures and methodology for obtaining valid conclusions is called:

- ► Descriptive Statistics
- ► Advance Statistics
- ► **Inferential Statistics (Page 237)**
- ► Sampled Statistics

Question No: 20 (Marks: 1) - Please choose one
Which of the following is a systematic arrangement of data into rows and columns?

- ► Classification
- ► **Tabulation**
- ► Bar chart
- ► Component bar chart

Question No: 21 (Marks: 1) - Please choose one
In normal distribution Q.D =

- ► 0.5σ
- ► 0.75σ
- ► 0.7979σ
- ► 0.6745σ Click here for detail

Question No: 22 (Marks: 1) - Please choose one
In normal distribution $\beta_2 =$

- ► 1
- ► 2
- ► **3 (Page 119)**
- ► 0

Question No: 23 (Marks: 1) - Please choose one
If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- ► Ogive
- ► Frequency polygon
- ► **Frequency curve (Page 38)**
- ► Historigram

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Question No: 24 (Marks: 1) - Please choose one
Which one of the following statements is true regarding a population?

- **It must be a large number of values (Page 12)**
- It must refer to people
- It is a collection of individuals, objects, or measurements
- It is small part of whole

Question No: 25 (Marks: 1) - Please choose one
When $Q_1 = 2$ *and* $Q_3 = 4$,what is the value of Median, if the distribution is symmetrical:

- 1
- **2**
- 3
- 4

Question No: 26 (Marks: 1) - Please choose one
In a simple linear regression model, if it is assumed that the intercept parameter is equal to zero, then:

- The regression line will pass through the origin
- The regression line will pass through the point (0,10).
- The regression line will pass through the point (0,-10).
- **The slope of the line will also be equal to 0.**

Question No: 27 (Marks: 1) - Please choose one
The degrees of freedom for a t-test with sample size 10 is:

- 5
- 8
- **9 (Page 298) rep**
- 10

Question No: 28 (Marks: 1) - Please choose one
In testing of hypothesis, we always begin it with assuming that:

- **Null hypothesis is true (Page 277) rep**
- Alternative hypothesis is true
- Sample size is large
- Population is normal

بدصورت چہرہ بدصورت دماغ سے بہتر ہے

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Question No: 29 (Marks: 1) - Please choose one
A failing student is passed by an examiner is an example of:

- Type I error
- **Type II error**
- Correct decision
- No information regarding student exams

Question No: 30 (Marks: 1) - Please choose one
How to find $P(X+Y \le 1)$?

- f(0, 0) + f(0, 1) + f(1, 2)
- f(2, 0) + f(0, 1) + f(1, 0)
- f(0, 0) + f(1, 1) + f(1, 0)
- f(0, 0) + f(0, 1) + f(1, 0)

FINAL TERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 3)

Question No: 1 (Marks: 1) - Please choose one
The total number of samples when sampling is done with replacement :

- N^n **(Page 237) rep**
- C_n^N
- $\frac{N-n}{N-1}$
-
- 1

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Question No: 2 (Marks: 1) - Please choose one
The test statistic used in analysis of variance procedure follow the ..... distribution.:

- ► χ^2
- ► T
- ► Z
- ► **F (Page 326) rep**

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If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- ► **0,1,2,3 rep**
- ► 1,3,3,1
- ► 1, 2, 3
- ► 3, 2

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If X and Y are independent variables, then E (XY) is:

- ► E(XX)
- ► **E(X).E(Y) (Page 202) rep**
- ► X.E(Y)
- ► Y.E(X)

Question No: 5 (Marks: 1) - Please choose one

$$S^2 = \frac{\sum(x-\bar{x})^2}{n}$$

The sample variance is:

- ► Unbiased estimator of σ^2
- ► **Biased estimator of σ^2 (Page 260) rep**
- ► Unbiased estimator of μ
- ► None of these

انسان دکھ نہیں دیتے بلکہ انسانوں سے وابستہ امیدیں دکھ دیتی ہیں

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Question No: 6 (Marks: 1) - Please choose one
When c is a constant, then E(c) is:

- ► 0
- ► 1
- ► **c (Page 180) rep**
- ► -c

Question No: 7 (Marks: 1) - Please choose one
When the random variable X and Y are independent, its co-variance is:

- ► One
- ► Negative
- ► **Zero (Page 207)**
- ► Positive

Question No: 8 (Marks: 1) - Please choose one
When f (x, y) is bivariate probability density function of continuous r.v.'s X and Y, then

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$$

is equal to:

- ► **1 (Page 199)**
- ► 0
- ► -1
- ► ∞

Question No: 9 (Marks: 1) - Please choose one
Which dispersion is calculated from all the observations?

- ► **Standard deviation**
- ► Quartile deviation
- ► Rang
- ► Coefficient of Rang

Question No: 10 (Marks: 1) - Please choose one
Standard deviation of the data 7, 7, 7, 7, 7, 7, 7 is

- ► 49
- ► Sqrt (7)
- ► **0 Standard deviation will always be zero if all the values in data are same**
- ► 7

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Question No: 11 (Marks: 1) - Please choose one
Which one is the poor measure of dispersion in open-end distribution?

- **Range**
- Standard deviation
- Mean deviation
- Variance

Question No: 12 (Marks: 1) - Please choose one
Men tend to marry women who are slightly younger than themselves. Suppose that every man married a woman who was exactly 5 years younger than themselves. Which of the following is correct:

- The correlation is −5
- The correlation is 5
- **The correlation is 1** Click here for detail
- The correlation is 0

Question No: 13 (Marks: 1) - Please choose one
Sum of absolute deviations of the values is least when deviations are taken from:

- **Mean**
- Median
- Mode
- Geometric mean

Question No: 14 (Marks: 1) - Please choose one
Which of the following measures of central location is affected most by extreme values:

- Geometric Mean
- Median
- **Mean** Click here for detail
- Mode

Question No: 15 (Marks: 1) - Please choose one
Which of the following is a critical value of Z when $1-\alpha = 95\%$ for one tailed test:

- 2.58
- 1.645
- 1.25
- **1.96**

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو

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Question No: 16 (Marks: 1) - Please choose one
In normal distribution Q.D =

- 0.5σ
- 0.75σ
- 0.7979σ
- 0.6745σ Click here for detail rep

Question No: 17 (Marks: 1) - Please choose one
The difference between expected value of statistic and parameter is called:

- Non-sampling error
- Sampling error
- Standard error
- **Bias** Click here for detail

Question No: 18 (Marks: 1) - Please choose one
In Statistics, we have MSE which is abbreviation of……

- **Mean square error (Page 330) rep**
- Measured square error
- Medical screening exam
- Major sampling error

Question No: 19 (Marks: 1) - Please choose one
The following data shows the number of hours worked by 200 statistics students.

Number of Hours	Frequency
0 - 9	40
10 - 19	50
20 - 29	70
30 - 39	40

What is its class interval?

- 9
- **10**
- 11
- Varies from class to class

خوبصورتی علم وادب سے ہوتی ہے لباس وحسن سے نہیں

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Question No: 20 (Marks: 1) - Please choose one
Which one is the formula of range:

- $x_m - x_0$
- $x_0 - x_m$
- $\frac{x_0 - x_m}{2}$
- $\frac{x_0 + x_m}{2}$ **(Page 80)**

Question No: 21 (Marks: 1) - Please choose one
Which one is the formula of Geometric mean for group data:

- $anti\log\left[\frac{\sum f \log X}{n}\right]$
- $anti\log\left[\frac{\sum \log X}{n}\right]$ **(Page 75)**
- $anti\log\left[\frac{\sum \log f X}{n}\right]$
- $anti\log\left[\frac{\sum \log f X}{\sum_{i=1}^{n} f}\right]$

Question No: 22 (Marks: 1) - Please choose one
The F-distribution has …… parameter.

- One
- No
- **Two (Page 312)**
- Three

تم اچھا کرو زمانہ تم کو برا سمجھے یہ اس سے بہتر ہے کہ تم برا کرو اور زمانہ تم کو اچھا سمجھے

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Question No: 23 (Marks: 1) - Please choose one

The sample mean $\bar{X} = \frac{\Sigma X}{n}$ is an unbiased estimator of population mean (μ), because:

- $E(\bar{X}) = 0$
- $E(\bar{X}) = \mu$ **(Page 259)**
- $E(\bar{X}) > \mu$
- $E(\bar{X}) < \mu$

Question No: 24 (Marks: 1) - Please choose one

For a particular hypothesis test, $\alpha = 0.05$ and $\beta = 0.10$. The power of test equals to:

- 0.95
- 0.25
- **0.90**
- 0.14

Question No: 25 (Marks: 1) - Please choose one

The degrees of freedom for a t-test with sample size 10 is:

- 5
- 8
- **9 (Page 298) rep**
- 10

Question No: 26 (Marks: 1) - Please choose one

The degrees of freedom for a t-test with sample size 14 is:

- 14
- **13 (Page 341) rep**
- 7
- 0

Question No: 27 (Marks: 1) - Please choose one

The degrees of freedom for a t-test with sample size 6 is:

- 1
- 3
- **5 (Page 341)**
- 7

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Question No: 28 (Marks: 1) - Please choose one
In testing of hypothesis, we always begin it with assuming that:

- **Null hypothesis is true (Page 277) rep**
- Alternative hypothesis is true
- Sample size is large
- Population is normal

Question No: 29 (Marks: 1) - Please choose one
If a continuous probability distribution has $\beta_2 = 2.14$ then what will be peakedness of the distribution?

- **Platykurtic (Page 119) rep**
- Mesokurtic
- Leptokutic
- Moderately skewed

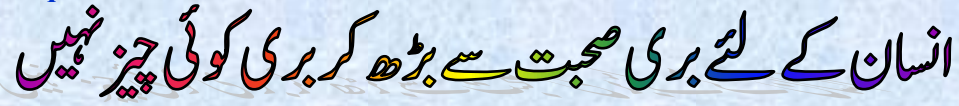
Question No: 30 (Marks: 1) - Please choose one
The difference between statistic and parameter is called:

- **Bias** Click here for detail **rep**
- Standard error
- Sampling error
- None of above

FINAL TERM EXAMINATION
Fall 2009
STA301- Statistics and Probability (Session - 4)

Question No: 1 (Marks: 1) - Please choose one
Mean deviation is always:

- Less than S.D
- Greater than S.D
- Greater or equal to S.D
- **Less or equal to S.D** Click here for detail



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Question No: 2 (Marks: 1) - Please choose one
The value of χ^2 can never be :

- Zero
- Less than 1
- Greater than 1
- **Negative (Page 307) rep**

Question No: 3 (Marks: 1) - Please choose one
The mean of the F-distribution is:

- $\frac{v_1}{v_1 - 2} \quad for v_1 \rangle 2$
- $\frac{v_2}{v_2 - 2} \quad for v_2 \rangle 2$ **(Page 312) rep**
- $\frac{v_1}{v_1 - 2} \quad for v_1 \geq 2$
- $\frac{v_2}{v_2 - 2} \quad for v_1 \leq 2$

Question No: 4 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

- $E(X)+E(Y)$
- $E(X)-E(Y)$ **(Page 202) rep**
- $X - E(Y)$
- $E(X)-Y$

خاموشی غصے کا بہترین علاج ہے

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Question No: 5 (Marks: 1) - Please choose one
Evaluate: (9-4)!

► 362880
► **120**
► 24
► 6

Question No: 6 (Marks: 1) - Please choose one
Which formula represents the probability of the complement of event A:

► 1 + *P* (*A*)
► **1 - *P* (*A*) (Page 156)**
► *P* (*A*)
► *P* (*A*) -1

Question No: 7 (Marks: 1) - Please choose one
Ideally the width of confidence interval should be:

► **0 (Page 270)**
► 1
► 99
► 100

Question No: 8 (Marks: 1) - Please choose one
If the sampling distribution of $\bar{X}$ is normal, the interval $\mu_{\bar{x}} \pm 3\sigma_{\bar{x}}$ includes:

► 99% of the sample means
► **99.73% of the sample means (Page 228)**
► 98% of the sample means
► 95% of the sample means

Question No: 9 (Marks: 1) - Please choose one
The probability distribution of a statistic is called the:

► Population distribution
► Frequency distribution
► **Sampling distribution** click here for detail
► Sample distribution

جھوٹ رزق کو کھا جاتا ہے

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Question No: 10 (Marks: 1) - Please choose one

An estimator T is said to be unbiased estimator of θ if

- **E (T) = θ (Page 258) rep**
- E (T) =T
- E (T) =0
- E (T) =1

Question No: 11 (Marks: 1) - Please choose one

If the following is a probability distribution, then what is the value of 'a':

X	1	2	3
P(X)	0.1	a	0.1

- 0.6
- **0.8**
- 0.2
- 0.4

Question No: 12 (Marks: 1) - Please choose one

A discrete probability function f(x) is always:

- Non-negative
- Negative
- **One (Page 168)**
- Zero

Question No: 13 (Marks: 1) - Please choose one

An expected value of a random variable is equal to:

- Variance
- **Mean (Page 191)**
- Standard deviation
- Covariance

افضل انسان وہ ہے جو اپنی اصلاح کی کوشش کرتا ہے

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Question No: 14 (Marks: 1) - Please choose one

$f(x|1) =$

The _____:

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$ **(Page 198)**
- $\frac{f(x,1)}{h(x)}$

Question No: 15 (Marks: 1) - Please choose one

The area under a normal curve between 0 and -1.75 is

- .0401
- .5500
- **.4599 (Page 230)**
- .9599

Question No: 16 (Marks: 1) - Please choose one

The continuity correction factor is used when:

- The sample size is at least 5
- Both *nP* and *n (1-P)* are at least 30
- **A continuous distribution is used to approximate a discrete distribution** Click here for detail
- The standard normal distribution is applied

Question No: 17 (Marks: 1) - Please choose one

Which of the following is impossible in sampling:

- Destructive tests
- **Heterogeneous**
- To make voters list
- None of these

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

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Question No: 18 (Marks: 1) - Please choose one
Which of the following is a systematic arrangement of data into rows and columns?

- ► Classification
- ► **Tabulation rep**
- ► Bar chart
- ► Component bar chart

Question No: 19 (Marks: 1) - Please choose one
Which one of the following statements is true regarding a sample?

- ► **It is a part of population (Page 13)**
- ► It must contain at least five observations
- ► It refers to descriptive statistics
- ► It produces True value

Question No: 20 (Marks: 1) - Please choose one
The data for an ogive is found in which distribution?

- ► A relative frequency distribution
- ► A frequency distribution
- ► A joint frequency distribution
- ► **A cumulative frequency distribution (Page 43)**

FINAL TERM EXAMINATION
Fall 2009
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
10! =……

- ► 362880
- ► **3628800**
- ► 362280
- ► 362800

اس سے پہلے کہ تمہیں شہوت فتنے میں ڈالے نکاح کرلو

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Question No: 2 (Marks: 1) - Please choose one
When E is an impossible event, then P(E) is:

► 2
► **0 (Page 146) rep**
► 0.5
► 1

Question No: 3 (Marks: 1) - Please choose one
The value of χ^2 can never be :

► Zero
► Less than 1
► Greater than 1
► **Negative (Page 307) rep**

Question No: 4 (Marks: 1) - Please choose one
The curve of the F- distribution depends upon:

► **Degrees of freedom (Page 312)**
► Sample size
► Mean
► Variance

Question No: 5 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

► $E(X)+E(Y)$
► $E(X)-E(Y)$ **(Page 202) rep**
► $X-E(Y)$
► $E(X)-Y$

Question No: 6 (Marks: 1) - Please choose one
In testing hypothesis, we always begin it with assuming that:

► **Null hypothesis is true (Page 277) rep**
► Alternative hypothesis is true
► Sample size is large
► Population is normal

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Question No: 7 (Marks: 1) - Please choose one

$$\frac{\ell^{-0.135}\ 0.135^{1}}{1!}$$

For the Poisson distribution P(x) = the mean value is :

► 2
► 5
► 10
► **0.135 (Page 222) rep**

Question No: 8 (Marks: 1) - Please choose one

When two coins are tossed simultaneously, P (one head) is:

► $\frac{1}{4}$
► $\frac{1}{2}$
► $\frac{3}{4}$
► 1

Question No: 9 (Marks: 1) - Please choose one

From point estimation, we always get:

► **Single value (Page 257) rep**
► Two values
► Range of values
► Zero

Question No: 10 (Marks: 1) - Please choose one

$$S^2 = \frac{\sum(x-\bar{x})^2}{n}$$

The sample variance is:

► Unbiased estimator of σ^2
► **Biased estimator of σ^2 (Page 260) rep**
► Unbiased estimator of μ
► None of these

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Question No: 11 (Marks: 1) - Please choose one
Var(4X + 5) =_____

► 16 Var (X)
► **16 Var (X) + 5 rep**
► 4 Var (X) + 5
► 12 Var (X)

Question No: 12 (Marks: 1) - Please choose one
When f (x, y) is bivariate probability density function of continuous r.v.'s X and Y, then

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$$

is equal to:

► **1 rep**
► 0
► -1
► ∞

Question No: 13 (Marks: 1) - Please choose one
The area under a normal curve between 0 and -1.75 is

► .0401
► .5500
► **.4599 (Page 230) rep**
► .9599

Question No: 14 (Marks: 1) - Please choose one
When a fair die is rolled, the sample space consists of:
► 2 outcomes
► 6 outcomes
► **36 outcomes (Page 145)**
►16 outcomes

Question No: 15 (Marks: 1) - Please choose one
When testing for independence in a contingency table with 3 rows and 4 columns, there are _____ degrees of freedom.
► 5
► **6 (Page 341)**
► 7
► 12

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Question No: 16 (Marks: 1) - Please choose one
The F- test statistic in one-way ANOVA is:

- ► SSW / SSE
- ► MSW / MSE
- ► SSE / SSW
- ► **MSE / MSW (Not sure)**

Question No: 17 (Marks: 1) - Please choose one
The continuity correction factor is used when:

- ► The sample size is at least 5
- ► Both *nP* and *n (1-P)* are at least 30
- ► **A continuous distribution is used to approximate a discrete distribution rep**
- ► The standard normal distribution is applied

Question No: 18 (Marks: 1) - Please choose one
A uniform distribution is defined by:

- ► **Its largest and smallest value** click here for detail
- ► Smallest value
- ► Largest value
- ► Mid value

Question No: 19 (Marks: 1) - Please choose one
Which graph is made by plotting the mid point and frequencies?

- ► **Frequency polygon (Page 34)**
- ► Ogive
- ► Histogram
- ► Frequency curve

Question No: 20 (Marks: 1) - Please choose one
In a set of 20 values all the values are 10, what is the value of median?

- ► 2
- ► 5
- ► **10**
- ► 20

ہر چیز کی ایک پہچان ہوتی ہے اور عقلمند کی پہچان غور و فکر کرنا ہے اور غور و فکر کی پہچان خاموشی ہے

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FINAL TERM EXAMINATION
Fall 2009
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
For a particular data the value of Pearson's coefficient of skewness is greater then zero. What will be the shape of distribution?

- ► **Negatively skewed**
- ► J-shaped
- ► Symmetrical
- ► **Positively skewed (Page 110)**

Question No: 2 (Marks: 1) - Please choose one
In measures of relative dispersion unit of measurement is:

- ► **Changed**
- ► **Vanish Rep**
- ► Does not changed
- ► Dependent

Question No: 3 (Marks: 1) - Please choose one
The F-distribution always ranges from:

- ► 0 to 1
- ► 0 to -∞
- ► -∞ to +∞
- ► **0 to +∞ (Page 312) rep**

Question No: 4 (Marks: 1) - Please choose one
In chi-square test of independence the degrees of freedom are:

- ► **n - p**
- ► n - p-1
- ► n - p- 2
- ► n – 2



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Question No: 5 (Marks: 1) - Please choose one
The Chi- Square distribution is continuous distribution ranging from:

- $-\infty \leq \chi^2 \leq \infty$
- $-\infty \leq \chi^2 \leq 1$
- $-\infty \leq \chi^2 \leq 0$
- **$0 \leq \chi^2 \leq \infty$ (Page 307) rep**

Question No: 6 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

- $E(X)+E(Y)$
- $E(X)-E(Y)$ **(Page 202) rep**
- $X-E(Y)$
- $E(X)-Y$

Question No: 7 (Marks: 1) - Please choose one
If ŷ is the predicted value for a given x-value and b is the y-intercept then the equation of a regression line for an independent variable x and a dependent variable y is:

- **ŷ = mx + b, where m = slope (Page 121) rep**
- x = ŷ + mb, where m = slope
- ŷ = x/m + b, where m = slope
- ŷ = x + mb, where m = slope

Question No: 8 (Marks: 1) - Please choose one
The location of the critical region depends upon:

- **Null hypothesis**
- **Alternative hypothesis (Page 281) rep**
- Value of alpha
- Value of test-statistic

وہ لوگ مبارک ہیں جو الفاظ سے نصیحت نہیں کرتے بلکہ عمل سے کرتے ہیں

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Question No: 9 (Marks: 1) - Please choose one
The variance of the t-distribution is give by the formula:

$$\sigma^2 = \sqrt{\frac{\nu}{\nu - 2}}$$

▶

$$\sigma^2 = \frac{\nu^2}{\nu - 2}$$

▶

$$\sigma^2 = \frac{\nu}{\nu - 1}$$

▶

$$\sigma^2 = \frac{\nu}{\nu - 2}$$ **(Page 293)**

▶

Question No: 10 (Marks: 1) - Please choose one
Which one is the correct formula for finding desired sample size?

$$n = \left(\frac{Z_{\alpha/2}.\sigma}{e} \right)^2$$ **(Page 276)**

▶

$$n = \left(\frac{Z_{\alpha/2}.\sqrt{\sigma}}{e} \right)^2$$

▶

$$n = \left(\frac{Z_{\alpha/2}.\bar{X}}{e} \right)^2$$

▶

$$n = \frac{Z_{\alpha/2}.\sigma}{e}$$

▶

خدا کے سوا کسی سے امید مت رکھو

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Question No: 11 (Marks: 1) - Please choose one
A discrete probability function f(x) is always:

- ► Non-negative
- ► Negative
- ► **One (Page 168) rep**
- ► Zero

Question No: 12 (Marks: 1) - Please choose one
E(4X + 5) =_____

- ► 12 E (X)
- ► **4 E (X) + 5**
- ► 16 E (X) + 5
- ► 16 E (X)

Question No: 13 (Marks: 1) - Please choose one
How P(X + Y < 1) can be find:

- ► f(0, 0) + f(0, 1) + f(1, 2)
- ► f(2, 0) + f(0, 1) + f(1, 0)
- ► f(0, 0) + f(1, 1) + f(1, 0)
- ► f(0, 0) + f(0, 1) + f(1, 0)

Question No: 14 (Marks: 1) - Please choose one
$f(x|1) =$
The _____:

- ► $f(1,1)$
- ► $f(x,1)$
- ► $\frac{f(x,1)}{h(1)}$ **(Page 198) rep**
- ► $\frac{f(x,1)}{h(x)}$
- ►

ہر کسی کی روٹی نہ کھا بلکہ ہر شخص کو اپنی روٹی کھلا

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Question No: 15 (Marks: 1) - Please choose one
The area under a normal curve between 0 and -1.75 is

- .0401
- .5500
- **.4599 (Page 230) rep**
- .9599

Question No: 16 (Marks: 1) - Please choose one
In normal distribution M.D. =

- 0.5σ
- 0.75σ
- 0.7979σ Click here for detail **rep**
- 0.6445σ

Question No: 17 (Marks: 1) - Please choose one
In an ANOVA test there are 5 observations in each of three treatments. The degrees of freedom in the numerator and denominator respectively are.....

- 2, 4
- 3, 15
- 3, 12
- **2, 12**

Question No: 18 (Marks: 1) - Please choose one
A set that contains all possible outcomes of a system is known as

- Finite Set
- Infinite Set
- **Universal Set (Page 134)**
- **No of these**

Question No: 19 (Marks: 1) - Please choose one
Stem and leaf is more informative when data is :

- Equal to 100
- Greater Than 100
- **Less than 100** click here for detail rep
- In all situations



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Question No: 20 (Marks: 1) - Please choose one
A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:

- ► Infinite population
- ► Finite population
- ► Concrete population
- ► **Hypothetical population (Page 12)**

STA 301 – Quizzes

Question # 1 of 10 (Total Marks: 1) Select correct option:
The number of telephone calls that pass through a switchboard has a Poison distribution with mean equal to 2 per minute. The probability that no telephone calls pass through the switchboard in two consecutive minutes is:

0.2707
0.0517
0.0183
0.0366

Question # 2 of 10 (Total Marks: 1) Select correct option:
The range of the binomial distribution is:
0, 1, 2, ... , 100
0, 1, 2, … , n Click here for detail
0, 1, 2, … , x
1, 2, … , n

Question # 3 of 10 (Total Marks: 1) Select correct option:
Which of the following is NOT CORRECT about a standard normal distribution?

$P(0 \leq Z \leq 1.50) = .4332$
$P(Z \geq 2.0) = .0228$
$P(Z \geq -2.5) = .4938$ (Page 230)
$P(Z \leq -1.0) = .1587$



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Question # 4 of 10 (Total Marks: 1) Select correct option:
The distribution function (df) is also known as cumulative distribution function (cdf).
Yes (Page 173)
No

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following pairs of events are mutually exclusive?

A:the numbers above100;B: the numbers less than-200
A:the odd numbers; B:the number 5
A:the even numbers; B:the numbers greater than 10
A:the numbers less than 5; B:all the negative numbers

Question # 6 of 10 (Total Marks: 1) Select correct option:
Two events A & B are said to be independent if...
P(A) + P(B)
P(B/A) = P(B)
P(A) * P(B) (Page 162)
P(A/B) = P(A)

Question # 7 of 10 (Total Marks: 1) Select correct option:
The collection of all outcomes for an experiment is called:

A sample space Click here for detail
Joint probability simple event
The intersection of events
Random experiment

Question # 8 of 10 (Total Marks: 1) Select correct option:
Symbolically, a conditional probability is:
P(AB)
P(A/B) (Page 159)
P(A)
P(AUB)

Question # 9 of 10 (Total Marks: 1) Select correct option:
Which of the following statements is INCORRECT about the sampling distribution of the sample mean?

The standard error of the sample mean will decrease as the sample size increases
The standard error of the sample mean is measure of the variability of the sample mean among repeated samples
The sample mean is unbiased for the true (unknown) population mean
The sampling distribution shows how the sample was distributed around the sample mean
Click here for detail

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Question # 10 of 10 (Total Marks: 1) Select correct option:
If one event is unaffected by the outcome of another event the two events are said to be:

1. Dependent
2. Not Mutually Exclusive
3. Mutually Exclusive

Independent Click here for detail

Question # 1 of 10 (Total Marks: 1) Select correct option:
Let X be a random variable with binomial distribution, that is (X=0,1,…, n). The expected value E[X] is

p
np (Page 214)
np(1-p)
Xnp

Question # 2 of 10 (Total Marks: 1) Select correct option:
The sample mean is an unbiased estimator for the population mean. This means:

The sample mean has a normal distribution
The average sample mean, over all possible samples, equals the population mean (Page 259)
The sample mean is always very close to the population mean
The sample mean will only vary a little from the population mean

Question # 4 of 10 (Total Marks: 1) Select correct option:
Probability of an impossible event is always:

Less than one
Greater than one
Between one and zero
Zero (Page 146)

Question # 5 of 10 (Total Marks: 1) Select correct option:
The function abbreviated to d.f. is also called the.....

Probability density function
Probability distribution function
Commutative distribution function (Page 173)
Discrete function



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Question # 6 of 10 (Total Marks: 1) Select correct option:
The total area under the normal curve is:

0
1 (Page 186)
0.5
0.75

Question # 7 of 10 (Total Marks: 1) Select correct option:
Two events A & B are said to be independent if....

P (A) + P (B)
P (B\A) = P (B)
P (A) * P (B) (Page 162) rep
P (A\B) = P (A)

Question # 8 of 10 (Total Marks: 1) Select correct option:
When two coins are tossed the probability of at most one head is:

1/4
2/4
3/4 rep
1

Question # 9 of 10 (Total Marks: 1) Select correct option:
For exhaustive events, the P(AUBUC) is equal to:

P(A)
P(S) (Page 147)
P(A) * P(B)* P(C)
P(B)

Question # 10 of 10 (Total Marks: 1) Select correct option:
One card is drawn from a standard 52 card deck. In describing the occurrence of two possible events, an Ace and a King, these two events are said to be:
Select correct option:
independent
randomly independent
random variables
mutually exclusive click here for detail

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STA 301 – Quizzes

Question # 1 of 10 (Total Marks: 1) Select correct option:
The number of parameters in hypergeometric distribution is (are):

1
2
3 (Page 219)
4

Question # 2 of 10 (Total Marks: 1) Select correct option:
If Y=bX, then variance of Y is

b*2 var(x)
var(x)
b var(x)
b square root var(x)

Question # 3 of 10 (Total Marks: 1) Select correct option:
If f(x) is a continuous probability function, then P(X = 2) is:

1
0 (Page 186)
1/2
2

Question # 4 of 10 (Total Marks: 1) Select correct option:
In regression line Y=a+bX, Y is called:

Dependent variable (Page 121)
Independent variable
Explanatory variable
Regressor

پہلے وقتوں میں تعلیم کم تھی اور علم زیادہ، آج کل علم کم ہے اور تعلیم زیادہ

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Question # 5 of 10 (Total Marks: 1) Select correct option:
If A and B are mutually exclusive events with P (A) =0.25 and P (B) = 0.50, Then P (A or B) =……

0.25
0.75 (Page 154)
0.50
1

Question # 6 of 10 (Total Marks: 1) Select correct option:
In a 52 well shuffled pack of 52 playing cards, the probability of drawing any one diamond card is

1/52
4/52
13/52
52/52

Question # 7 of 10 (Total Marks: 1) Select correct option:
Probability of a sure event is

8
1 (Page 154)
0
0.5

Question # 8 of 10 (Total Marks: 1) Select correct option:
If Y=3X+5,then S.D of Y is equal to

9 s.d(x)
3 s.d(x)
s.d(x)+5
3s.d(x)+5

Question # 9 of 10 (Total Marks: 1) Select correct option:
The probability of drawing a red queen card from well-shuffled pack of 52 playing cards is

4/52
2/52
13/52
26/52

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Question # 10 of 10 (Total Marks: 1) Select correct option:
If P (B|A) = 0.25 and P (A and B) =0.20, then P (A) is

0.05
0.80 (Page 159)
0.95
0.75

Question # 1 of 10 (Total Marks: 1) Select correct option:
A student solved 25 questions from first 50 questions of a book to be solved. The probability that he will solve the remaining all questions is:

0.25
0.5
1
0

Question # 1 of 10 (Total Marks: 1) Select correct option:
The classical definition of probability assumes:

Exhaustive events
Mutually exclusive events
Equally likely evens (Page 151)
Independent evens

Question # 2 of 10 (Total Marks: 1) Select correct option:
In scatter diagram, the variable plotted along Y-axis is:
Independent variable
Dependent variable
Continuous variable
Discrete variable

Question # 3 of 10 (Total Marks: 1) Select correct option:
Which of the following measures of dispersion are based on deviations from the mean?

Variance
Standard deviation
Mean deviation
All of the these

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Question # 4 of 10 (Total Marks: 1) Select correct option:
What does it mean when a data set has a standard deviation equal to zero?
All values of the data appear with the same frequency.
The mean of the data is also zero.
All of the data have the same value. Click here for detail
There are no data to begin with.

Question # 5 of 10 (Total Marks: 1) Select correct option:
A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as _____.

Probability distribution
The expected return
The standard deviation
Coefficient of variation

Question # 6 of 10 (Total Marks: 1) Select correct option:
Which of the following can never be probability of an event?

0
1
0.5
-0.5

Question # 7 of 10 (Total Marks: 1) Select correct option:
The standard deviation of -1, -1, -1, -1 will be

1
-1
0
Does not exist

Question # 8 of 10 (Total Marks: 1) Select correct option:
The Special Rule of Addition is used to combine:

Independent Events
Mutually Exclusive Events click here for detail
Events that total more than 1.00
Events based on subjective probabilities

بہادر آدمی ایک مرتبہ جبکہ بزدل آدمی کئی مرتبہ مرتا ہے

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Question # 9 of 10 (Total Marks: 1) Select correct option:
set which is the sub-set of every set is

Empty Set (Page 134)
Power Set
Universal Set
Super Set

Question # 10 of 10 (Total Marks: 1) Select correct option:
When two dice are rolled the number of possible sample points is :

6
12
24
36 (Page 145)

Question # 1 of 10 (Total Marks: 1) Select correct option:
If two events A and B are not mutually exclusive then

P (A or B) = P (A) + P (B) – P (A and B) (Page 157)
P (A or B) = P (A) + P (B)
P (A or B) = P (A) x P (B)
P (A or B) = P (A) + P (B)

Question # 2 of 10 (Total Marks: 1) Select correct option:
Evaluate (10-4)!

1000
720
480
32

Question # 3 of 10 (Total Marks: 1) Select correct option:
When we toss a coin , we get only:

1 outcome
2 outcome
3 outcome
4 outcome



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Question # 4 of 10 (Total Marks: 1) Select correct option:
If we roll a die then probability of getting a '6' will be

2/6
1/6
4/6
1

Question # 5 of 10 (Total Marks: 1) Select correct option:
If P(A) = 0.45, P(B) = 0.35, and P(A and B) = 0.25, then P(A | B) is:

1.4
1.8
0.714 0.25/0.35 = 0.714 (Page 159)
0.556

Question # 6 of 10 (Total Marks: 1) Select correct option:
Which of the following is not a measure of central tendency?

Percentile
Quartile
Standard deviation
Mode

Question # 7 of 10 (Total Marks: 1) Select correct option:
Random experiment can be repeated any no. of times under the…… conditions.
Different
Similar (Page 144)

Question # 9 of 10 (Total Marks: 1) Select correct option:
The simultaneous occurrence of two events is called:
Joint probability (Page 194)
Subjective probability
Prior probability
Conditional probability

Question # 10 of 10 (Total Marks: 1) Select correct option:
In regression analysis, the variable that is being predicted is the

Dependent variable click here for detail
Independent variable
Intervening variable
None of these

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Question # 1 of 10 (Total Marks: 1) Select correct option:
The probability of continuous random variable x on any particular point is always zero..
Yes (Page 186)
No

Question # 2 of 10 (Total Marks: 1) Select correct option:
P(an event) =no of favorable outcome/total no. of outcomes is the definition of
Subjective approach (Page 149)
Objective approach

Question # 3 of 10 (Total Marks: 1) Select correct option:
If C is a constant ,then E(c)=
0
1
C (Page 180) rep
-c

Question # 4 of 10 (Total Marks: 1) Select correct option:
When we toss a fair coin 4 times, the sample space consists of….points.
4
8
12
16

Question # 5 of 10 (Total Marks: 1) Select correct option:
If we roll three fair dices then the total number of outcomes is:
6
36
216 6*6*6=216
1296

Question # 6 of 10 (Total Marks: 1) Select correct option:
The probability of an event is always:
greater than 0
less than 1
between o and 1 click here for detail
greater than 1

Question # 7 of 10 (Total Marks: 1) Select correct option:
In a multiplication theorem P (A and B) equals:
P(A)
P (B) P(A) + P (B)
P(A) * P (B|A) (Page 159)
P(B\A)*P(B)

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Question # 8 of 10 (Total Marks: 1) Select correct option:
If a die is rolled, what is the probability of getting an even number greater than 2?
1/2
1/3 2/6 = 1/3
2/3
5/6

Question # 9 of 10 (Total Marks: 1) Select correct option:
In a Discrete probability distribution, P (x > 23) is read as:
P (there are more than 23 successes)
P (there are less than 23 successes)
P (there are at least 23 successes)
P (there are at most 23 successes)

Question # 10 of 10 (Total Marks: 1) Select correct option:
A dormitory on campus houses 200 students. 120 are male, 50 are upper division students, and 40 are upper division male students.A student is selected at random. The probability of selecting a lower division student, given the student is a female, is:

7/8
7/20
7/15
1/4

Question # 1 of 10 (Total Marks: 1) Select correct option:
The function F(x) gives the probability of the event that X takes a value .....
Less than x
Greater or equal to x
Less or equal x (Page 173)
Equal to x

Question # 2 of 10 (Total Marks: 1) Select correct option:
In a simple regression line model ,it is assume that the intercept parameter is equal to zero,
The regression line will pass through the origin.
The regression line will pass through the point (0,10)
The regression line will pass through the point (0,-10)
The slope of the line will also be zero.

Question # 3 of 10 (Total Marks: 1) Select correct option:
If p(AnB)=p(A/B).p(B),then A and B are
Independent
Dependant
Equally likely
Mutually exclusively

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Question # 4 of 10 (Total Marks: 1) Select correct option:
A fair coin is tossed three times, the probability that at least one head appears,

1/8
7/8 (HHH,HHT,HTH, HTT, THH THT TTH TTT)
3/8
5/8

Question # 5 of 10 (Total Marks: 1) Select correct option:
In probability distribution, the sum of probabilities is equals to

0
0.1
0.5
1 click here for detail

Question # 6 of 10 (Total Marks: 1) Select correct option:
When a coin is tossed 3 times, the probability of getting 3 tails is

1/8 (HHH,HHT,HTH, HTT, THH THT TTH TTT)
3/8
3/6
2/8

Question # 7 of 10 (Total Marks: 1) Select correct option:
In how many ways can a team of 11 players be chosen from a total of 16 players?

4368
2426
5400
2680

Question # 8 of 10 (Total Marks: 1) Select correct option:
The standard deviation of c (constant) is

c
c square
0 click here for detail
does not exist

اچھائی کرنے کے لئے ہمیشہ کسی بہانے کی تلاش میں رہو

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Question # 9 of 10 (Total Marks: 1) Select correct option:
Let E and F be events associated with the same experiment. Suppose the E and F are independent and that P(E) = 1/4 and P(F) = 1/2 Then P(E U F) is:

1/8
3/4 (Page 157)
7/8
5/8

STA 301 – Quizzes

Question # 1 of 10 (Total Marks: 1) Select correct option:
In which of the following situations binomial distributions is approximate to normal distribution?

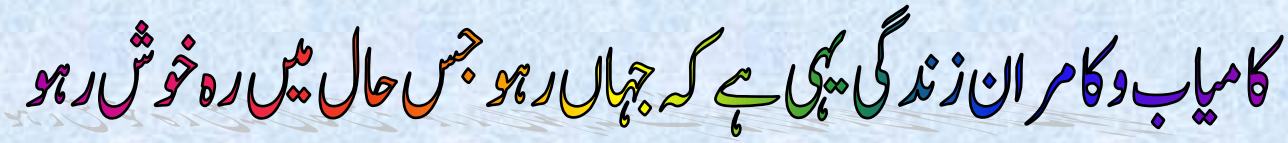
n = 50, p = 0.01
n = 500, p = 0.001
n = 100, p = 0.05
n = 50, p = 0.02 click here for detail

Question # 2 of 10 (Total Marks: 1) Select correct option:
The location and shape of the normal curve is (are) determined by:

Mean
Variance
Mean & variance
Mean & standard deviation click here for detail

Question # 3 of 10 (Total Marks: 1) Select correct option:
A random experiment has five outcomes in its sample space {s1, s2, s3, s4, s5}. If P(s1)=0.2,P(s2)=0.3, P(s3)=0.1 and P(s4)=0.2 then P(s5)=?

1
0.2
0.8
0.5



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Question # 4 of 10 (Total Marks: 1) Select correct option:
The joint density function f(x,y) will be a pdf if

Both of integrals f(x,y) dx dy =1 (Page 200)
Both of integrals f(x,y) dx dy >1
Both of integrals f(x,y) dx dy <1
Both of integrals f(x,y) dx dy =0

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following is correct property for joint probability distribution of X and Y:

Sigma f(X,Y)=1 (Page 194)
Sigma f(Y,X)=1
Both of above
None of above

Question # 6 of 10 (Total Marks: 1) Select correct option:
A random variable that can assume every possible value in an interval [a, b]:

Discrete variable
Continuous variable (Page 9)
Qualitative variable
Categorical variable

Question # 7 of 10 (Total Marks: 1) Select correct option:
Normal approximation to the binomial distribution is used when:

np>5
nq>5
Both of the above (Page 235)
None of the above

Question # 8 of 10 (Total Marks: 1) Select correct option:
The Maximum Likelihood Estimators (MLE) are …… and …… but not necessarily ……
Unbiased, consistent, efficient
Consistent, unbiased, efficient
Unbiased, efficient, consistent
Consistent, efficient, unbiased (Page 264)

کتنا شریف ہے وہ غمزدہ دل جو سب کو خوش کرنے کی کوشش کرتا ہے

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Question # 9 of 10 (Total Marks: 1) Select correct option:
A probability density function 'f(x)' has the following property:

f(x) <= 0
f(x) < 0
f(x) > 0 (Page 186)
f(x) >= 0

Question # 10 of 10 (Total Marks: 1) Select correct option:
For a continuous random variable X, P(X = x) is:
Select correct option:

0 (Page 186)
0.5
1
undefined

Question # 1 of 10 (Total Marks: 1) Select correct option:
An urn contains 4 red balls and 6 green balls. A sample of 4 balls is selected from the urn without replacement. It is the example of:

Binomial distribution
Hypergrometric distribution (Page 219)
Poisson distribution
Exponential distribution

Question # 2 of 10 (Total Marks: 1) Select correct option:
The probability distribution of the proportion of successes in all possible samples is called the:

Sampling distribution (Page 237)
Sampling probability distribution
Sampling distribution of sample proportions
Sampling distribution of Population proportions

Question # 3 of 10 (Total Marks: 1) Select correct option:
For good approximation of Poisson distribution to the binomial distribution, which of the following condition (s) is/are required:

The population size is large relative to the sample size
The probability, p, is close to .5 and the population size is large
The probability, p, is small and the sample size is large (Page 222)
The probability, p, is close to .5 and the sample size is large

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Question # 4 of 10 (Total Marks: 1) Select correct option:
If an estimator is more efficient then the other estimator, its shape of the sampling distribution will be

Flattered
Normal
Highly peaked (Page 261)
Skewed to right

Question # 6 of 10 (Total Marks: 1) Select correct option:
Match the binomial probability P(x< 23) with the correct statement.

P(there are at most 23 successes)
P (there are fewer than 23 successes)
P (there are more than 23 successes)
P (there are at least 23 successes)

Question # 7 of 10 (Total Marks: 1) Select correct option:
In statistics, the term 'expected value' implies the ___ value.

Independent
Normal
Standard
Mean (Page 191)

Question # 8 of 10 (Total Marks: 1) Select correct option:
A quantity obtained by applying certain rule or formula is known as

Estimate (Page 264)
Estimator
Parameter
Proportion

Question # 9 of 10 (Total Marks: 1) Select correct option:
Total no. of possible samples of size 2 (without replacement) from the population of size 6, will be:

20
15
10
18

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Question # 10 of 10 (Total Marks: 1) Select correct option:
When a coin is tossed 3 times, the probability of getting 3 or less tails is

1/2 (not sure)
0
1
3/5

Question # 1 of 10 (Total Marks: 1) Select correct option:
Total no. of possible samples of size 3 (with replacement) from the population of size 6, will be:

256
196
216 **6^3**
325

Question # 2 of 10 (Total Marks: 1) Select correct option:
Using the normal approximation to the binomial distribution with n= 3 and p= 0.0571 the value of mean is:

0.1713 **(Page 235)**
0.2132
0.5133
0.1923

Question # 3 of 10 (Total Marks: 1) Select correct option:
Suppose 60% of a herd of cattle is infected with a particular disease. Let Y = the number of non-diseased cattle in a sample of size 5. the distribution of Y is:

Binomial with n = 5 and p = 0.6
Binomial with n = 5 and p = 0.4
Binomial with n = 5 and p = 0.5
Poisson with u = .6

Question # 4 of 10 (Total Marks: 1) Select correct option:
The probability of success changes from trial to trial, is the property of:

Binomial experiment (Page 211)
Hypergeometric experiment
Both binomial & hypergeometric experiment
Poisson experiment

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Question # 5 of 10 (Total Marks: 1) Select correct option:
If a statistic used as an estimator, has its expected value equal to the true value of the population parameter being estimated then it is called.....

Consistent
Unbiased (Page 258)
Efficient
Sufficient

Question # 6 of 10 (Total Marks: 1) Select correct option:
In moments method, how many equations are needed to find the 2 unknown parameters?

2 (Page 263)
3
n/2

Question # 7 of 10 (Total Marks: 1) Select correct option:
Which of the following is desirable for a good point estimator?

Consistency
Unbiasedness
Efficiency
All of these (Page 258)

Question # 8 of 10 (Total Marks: 1) Select correct option:
The distribution function (df) is also known as

Probability distribution
Probability mass function
Probability density function
Cumulative distribution function (Page 173)

Question # 9 of 10 (Total Marks: 1) Select correct option:
If X and Y are two discrete r.v's with joint probability function f(x,y), then the conditional probability function Y given X, f(y|x) is given by

f(yj | xi) = f(xi, yj) / g(xi) (Page 195)
f(yj | xi) = f(xi, yj) / h(yj)
f(yj | xi) = f(xi, yj) / Sum of g(xi)
f(yj | xi) = f(xi, yj) / sum of h(yj)

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Question # 1 of 10 (Total Marks: 1) Select correct option:
Which of the probability distributions has three parameters?

Binomial distribution
Normal distribution
Hypergeometric distribution (Page 219)
Poisson distribution

Question # 2 of 10 (Total Marks: 1) Select correct option:
A standard deviation obtained from sampling distribution of sample statistics is known as

Sampling error
Standard error (Page 240)
Minimum error
Universal error

Question # 3 of 10 (Total Marks: 1) Select correct option:
A parameter is a ..... quantity.

Constant
Variable
Sample
Random

Question # 4 of 10 (Total Marks: 1) Select correct option:
How can you define statistical inference?

A decision, estimate, prediction or generalization about the population based on information contained in a sample click here for detail

A statement made about a sample based on the measurements in that sample
A set of data selected from a larger set of data
A decision, estimate, prediction or generalization about sample based on information contained in a population

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following is most important and most widely used method in point estimation?

The method of moments
The method of fractional moments
The method of leas square (Page 263)
The method of maximum likelihood

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Question # 6 of 10 (Total Marks: 1) Select correct option:
Binomial distribution is skewed to the right if:

p=q
P<q (Page 215)
p>q
p=n

Question # 7 of 10 (Total Marks: 1) Select correct option:
In a discrete distribution function, F(23) can be stated as:

P (there are at most 23 successes)
P (there are at least 23 successes)
P (there are less than 23 successes)
P (there are more than 23 successes)

Question # 8 of 10 (Total Marks: 1) Select correct option:
The Probabilities of the various values of the sample satistic can be computed using the..... definition of probability.

Subjective
Objective
Classical (Page 149)
None of the baove

Question # 9 of 10 (Total Marks: 1) Select correct option:
E(10X +3) =_____

10 E(X)
E(X)+3
10 E(X)+3
100E(X)

Question # 10 of 10 (Total Marks: 1) Select correct option:
If p is very small and n is considerably large then we shall apply the:

Binomial distribution
Hypergeometric distribution
Poisson distribution (Page 222)
Exponential distribution

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Question # 1 of 10 (Total Marks: 1) Select correct option:
Which of the following is NOT applicable to a Poisson distribution?

IF P = 0.5 & n = 19
IF P = 0.01 & n = 200
IF P = 0.02 & n = 300
IF P = 0.03 & n = 500

Question # 2 of 10 (Total Marks: 1) Select correct option:
The normal distribution has points of infection which are equidistance from the:

Median
Mean (Page 229)
Mode
Mean,Median & Mode

Question # 3 of 10 (Total Marks: 1) Select correct option:
If a random variable X denotes the number of heads when we toss a fair coin 5 times, the X assumed the values:

0,1,2,3
1, 2,3,4,5
0, 1, 2,3,4,5
1, 5, 5

Question # 4 of 10 (Total Marks: 1) Select correct option:
As a rule of thumb, when n>=30, then we can assume that……is normally distributed:

Probability distribution
Sampling distribution (Page 243)
Binomial distribution
Sampling distribution of sample mean

Question # 5 of 10 (Total Marks: 1) Select correct option:
If b(x, 7, 0.30), the variance of this distribution is:

1.77
1.74
1.44
1.47 (Page 333)

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Question # 6 of 10 (Total Marks: 1) Select correct option:
Which of the following is a characteristic of a binomial probability experiment?

Each trial has more than two possible outcomes
P(success) = P(failure)
Probability of success changes for each trail
The result of one trial does not affect the probability of success on any other trial (Page 211)

Question # 7 of 10 (Total Marks: 1) Select correct option:
The number of parameters in uniform distribution is (are):

1
2 (Page 225)
3
4

Question # 8 of 10 (Total Marks: 1) Select correct option:
The probability can never be:

1
1/2
1
-1/2

Question # 9 of 10 (Total Marks: 1) Select correct option:
The conditional probability P (A\B) is:

P(A n B)/P(B) (Page 159)
P(A n B)/P(A)
P(A U B)/P(B)
P(A U B)/P(A)

Question # 10 of 10 (Total Marks: 1) Select correct option:
A random sample of n=25 values gives sample mean 83. Can this sample be regarded as drawn from a normal population with μ= 80 and s= 7? In this question the alternative hypothesis will be:

H1: μ = 80
H1: μ ≠ 80
H1: μ > 80
H1: μ <80

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Question # 1 of 10 (Total Marks: 1) Select correct option:
The binomial distribution is negatively skewed when:
p>q (Page 215)
p<q
p=q
p=q=1/2

Question # 2 of 10 (Total Marks: 1) Select correct option:
When we draw the sample with replacement, the probability distribution to be used is:
Binomial
Hypergeometric
Binomial & hypergeometric
Poisson

Question # 3 of 10 (Total Marks: 1) Select correct option:
The moment ratios of normal distribution come out to be:
0 and 1
0 and 2
0 and 3 (Page 227)
0 and 4

Question # 4 of 10 (Total Marks: 1) Select correct option:
Suppose the test scores of 600 students are normally distributed with a mean of 76 and standard deviation of 8. The number of students scoring between 70 and 82 is:
272
164
260
328 click here for detail

Question # 5 of 10 (Total Marks: 1) Select correct option:
If P (A) = 0.3 and P (B) = 0.5, find P (A/B) where 'A' and 'B' are independent:

0.3
0.5
0.8
0.15 (Page 162)

Question # 6 of 10 (Total Marks: 1) Select correct option:
If the second moment ratio is less than 3 the distribution will be:
Mesokurtic
Leptokurtic
Platykurtic
None of these

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Question # 7 of 10 (Total Marks: 1) Select correct option:
For the independent events A and B if P (A) = 0.25, P (B) =0.40 then P (A and B) =……
Select correct option:

0.65
0.1 (Page 162)
0.50
0.15

Question # 8 of 10 (Total Marks: 1) Select correct option:
A random variable X has a probability distribution as follows: X | 0 1 2 3 P(X) | 2k 3k 13k 2k What is the possible value of k:

0.01
0.03
0.05
0.07

Question # 9 of 10 (Total Marks: 1) Select correct option:
The probability of drawing any one spade card is:
1/52
4/52
13/52
52/52

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STA301- Statistics and Probability

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1- A sample of size n=3 drawn without replacement experiment from a population N=5 items whose values 0,2,3,6,7 draw possible samples

Answer:-

Sample without replacement = $\binom{N}{n} = \binom{5}{3} = 10$

(0,2,3), (0,2,6), (0,2,7), (0,3,6), (0,3,7), (0,6,7), (2,3,6), (2,3,7), (2,6,7), (3,6,7).

2- Calculate sampling error if sample mean is 102 and population mean is 100

Answer:- (Page 14)

Sampling error $= \overline{X} - \mu$

$= 102 - 100$

$= 2$

3- Define Unbiased estimator

Answer:- (Page 258)

An estimator is defined to be unbiased if the statistic used as an estimator has its expected value equal to the true value of the population parameter being estimated. $E(\hat{\theta}) = \theta$

4- Quartile deviation

Answer:- (Page 84)

The quartile deviation is defined as half of the difference between the third and first quartiles.

$$Q.D = \frac{Q_3 - Q_1}{2}$$

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے

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Q1.decribe about significance level

Answer:

The significance level is the criterion used for rejecting the null hypothesis. It show you how likely a result is due to chance. The most frequently used values of α, the significance level, are 0.05 and 0.01, i.e. 5 percent and 1 percent.

Q2 .discuss about center limit theorem

Answer:- (Page 243)

"If a variable X from a population has mean μ and finite variance σ/2, then the sampling distribution of the sample mean−X approaches a normal distribution with mean μ and variance σ2/n as the sample size n approaches infinity.**"**

Q3. What is difference between constant and random variable

Answer:

A variable whose value cannot be changed once it has been assigned a value while those variable whose values changes when we assign value to it.

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Q: Define an Un-Biased estimator? 2 Marks

Answer: Rep

Q: Briefly Explain an experiment design? 2 Marks

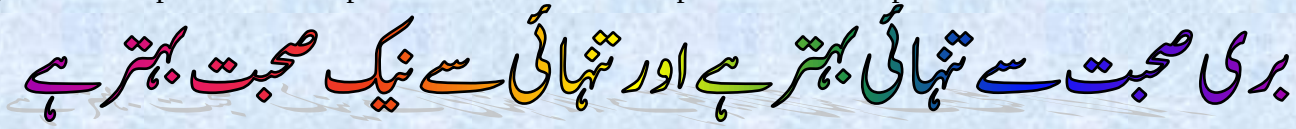
Answer:- (Page 320)

By an experimental design, we mean a plan used to collect the data relevant to the problem under study in such a way as to provide a basis for valid and objective inference about the stated problem.

Q: Define an independent and dependent variable in regression? 2 Marks

Answer:- (Page 121)

In regression Y represents the dependent variable and X represents the independent variable



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Find the mean and variance for the sampling distribution given below. 5 Marks

$(\hat{p})$	No. of Samples	Probability $f(\hat{p})$
0	1	1/20
1/3	9	9/20
2/3	9	9/20
1	1	1/20
Σ	20	1

Answer:- (Page 247)

$\hat{p}$	No. of samples	$f(\hat{p})$	$\hat{p}.f(\hat{p})$	$\hat{p}^2.f(\hat{p})$
0	1	$\frac{1}{20}$	0	0
$\frac{1}{3}$	9	$\frac{9}{20}$	$\frac{3}{20}$	$\frac{1}{20}$
$\frac{2}{3}$	9	$\frac{9}{20}$	$\frac{3}{10}$	$\frac{1}{5}$
1	1	$\frac{1}{20}$	$\frac{1}{20}$	$\frac{1}{20}$
$\sum$	20	1	$\sum \hat{p}.f(\hat{p}) = \frac{1}{2}$	$\sum \hat{p}^2.f(\hat{p}) = \frac{3}{10}$

$$\mu_{\hat{p}} = \sum \hat{p}.f(\hat{p}) = \frac{1}{2} = 0.5$$

$$\sigma^2_{\hat{p}} = \sum \hat{p}^2.f(\hat{p}) - \left[\sum \hat{p}.f(\hat{p})\right]^2$$

$$= \frac{3}{10} - \left[\frac{1}{2}\right]^2$$

$$= \frac{3}{10} - \frac{1}{4}$$

$$= \frac{1}{20} = 0.05$$



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What is the difference between an outcome and an event? (2)

Answer:- (Page 145)

An outcome is a result of a single trial of an experiment while an event is an individual outcome or any number of outcomes.

The mean of a Poisson distribution is 5 while its standard deviation is 4. Comment on it (2)

Answer:- (Page 223)

In Poisson distribution mean and variance are always equal but this statement is not satisfying this property of Poisson distribution.

If an automobile is driven on the average no more than 16000 Km per year, then formulate the null and alternative hypothesis. (2)

Answer:

$H \leq 16000$

$H1 > 16000$

Q: Discuss three properties of normal distribution? 3 Marks

Answer:- (Page 227)

Normal distribution is absolutely symmetrical, hence , μ_3 the third moment about the mean is zero

The normal curve is asymptotic to the x-axis as $x \to \pm \infty$.

For the normal distribution, it can be mathematically proved that $\mu_4 = 3\sigma_4$

Q: The 90% confidence interval for the population mean is 11 to 20, interpret this result? 3 Marks

Answer: - We are 90% sure that our Population mean lie between 11 – 20.

Q: Define LSD test? 3 Marks

Answer:- (Page 330)

According to this procedure, we compute the smallest difference that would be judged significant, and compare the absolute values of all differences of means with it. This smallest difference is called the least significant difference or

LSD. And is given by

$$LSD = t_{\frac{\alpha}{2},(v)} \sqrt{\frac{2(MSE)}{r}}$$

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

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How many parameters are associated with F- distribution and what is the range of the distribution? (3)
Answer:- (Page 312)
It has two parameters v1 and v2 which are known as the degrees of freedom and it ranging from zero to plus infinity.

Which of the following statement represents continuous data and discrete data? (5)
i) Number of shopes in a plaza. **Discrete data**
ii) Hourly temperature recorded by whether bureau. **Continuous data**
iii) Inches of rainfall in a city. **Continuous data**
iv) Number of passengers carried by rail every year. **Discrete data**
v) Height measurements of boys studying in a college. **Discrete data**

If the population proportions are gives as: P1 = 0.4, P2 = 0.20
find sigma^2 P-hat 1 - P-hat 2 , where n = 12.
Answer:- (Page 256)

$$\sigma_{\hat{p}1-\hat{p}2} = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}$$

$$q_1 = 1 - p_1 \Rightarrow 1 - 0.40 = 0.6$$

$$q_2 = 1 - p_2 \Rightarrow 1 - 0.20 = 0.8$$

$$\sigma_{\hat{p}1-\hat{p}2} = \frac{(0.4)(0.6)}{10} + \frac{(0.20)(0.80)}{10}$$

$$= 0.024 + 0.016$$

$$= 0.04$$

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what are steps involved in statistical research
Answer: (Page 11)
STEPS INVOLVED IN ANY STATISTICAL RESEARCH
• Topic and significance of the study
• Objective of your study
• Methodology for data-collection
 Source of your data
 Sampling methodology
 Instrument for collecting data

زندگی میں کامیابی کا یہی راز ہے کہ پریشانیوں سے پریشان مت بنو

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Q. (Marks 5)Two different dice are thrown. Make the sample space and also find the number of sample points in an even A that the sum is 7.

Answer:-

$$S = \begin{pmatrix} (1,1) & (2,1) & (3,1) & (4,1) & (5,1) & (6,1) \\ (1,2) & (2,2) & (3,2) & (4,2) & (5,2) & (6,2) \\ (1,3) & (2,3) & (3,3) & (4,3) & (5,3) & (6,3) \\ (1,4) & (2,4) & (3,4) & (4,4) & (5,4) & (6,4) \\ (1,5) & (2,5) & (3,5) & (4,5) & (5,5) & (6,5) \\ (1,6) & (2,6) & (3,6) & (4,6) & (5,6) & (6,6) \end{pmatrix}$$

$$A = \{(1,6) \quad (2,5) \quad (3,4) \quad (4,3) \quad (5,2) \quad (6,1)\}$$

Q. (Marks 2) Define the error sample and also write its formula.

Answer:- (Page 13)

The difference between the estimate derived from the sample (i.e. the statistic) and the true population value (i.e. the
parameter) is technically called the sampling error

Sampling error $= \overline{X} - \mu$

Q. (Marks 2) Properties of hyper geometric distribution.

Answer:- (Page 218)

- The outcomes of each trial may be classified into one of two categories, success and failure.
- The probability of success changes on each trial.
- The successive trials are not independent.
- The experiment is repeated a fixed number of times

5 number summery

Answer:- (Page 97)

A five-number summary consists of X0,Q1, Median, Q3, and Xm ; It provides us quite a good idea about the shape of the distribution.



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1.what s bias?(2)

Answer:- (Page 258)

$E(\hat{\theta}) \neq \theta$, the statistic is said to be a biased estimator

2.advantages and disadvantages of median?(3)

Answer:- Click here for detail

Advantages

(1) It is very simple to understand and easy to calculate. In some cases it is obtained simply by inspection.
(2) Median lies at the middle part of the series and hence it is not affected by the extreme values.
(3) In grouped frequency distribution it can be graphically located by drawing ogives.
(4) It is especially useful in open-ended distributions.

Disadvantages

(1) In simple series, the item values have to be arranged. If the series contains large number of items, then the process becomes tedious.
(2) It is a less representative average because it does not depend on all the items in the series.

3. Mathematical expectation of discrete random variable?(3)

Answer:- (Page 179)

In probability theory the expected value (or mathematical expectation) of a random variable is the sum of the product of the values within the range of the discrete random variable and their respective probabilities of occurrence.

$$E(X) = \sum_{i=1}^{n} x_i f(x_i)$$

4.any two properties of mathematical expectation?(2)

Answer:- (Page 202)

The important properties of the expected values of a random variable are as follows:

• If c is a constant, then E(c) = c. Thus the expected value of a constant is constant itself. This point can be understood easily by considering the following interesting example: Suppose that a very difficult test was given to students by a professor, and that every student obtained 2 marks out of 20! It is obvious that the mean mark is also 2. Since the variable 'marks' was a constant, therefore its expected value was equal to itself.

• If X is a discrete random variable and if a and b are constants, then E(aX + b) = a E(X) + b.

5.what s statistical test?(2)

Answer:- (Page 279)

A statistic, which provides a basis for testing a null hypothesis, is called a test-statistic. Every test-statistic has a probability distribution (i.e. sampling distribution) which gives the probability that our test-statistic will assume a value greater than or equal to a specified value OR a value less than or equal to a specified value when the null hypothesis is true.

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6. Decide a small sample and large sample?(2)

Answer:-

If the sample size 'n' is less than or equal to 30 (<=30) then it is said to be small sample, and it the sample size is larger than 30 then it is said to be large population.

10.tell the null and alternative hypothesis of 150?(2)

Answer:-

$H_0 = \mu = 150$

$H_1 = \mu \neq 150$

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35. In which condition, Poisson distribution is used to approximate the hyper geometric distribution?

Answer:- (Page 224)

The Poisson distribution can be used to approximate the hyper geometric distribution when n < 0.05N, n > 20, and p < 0.05

36. Elaborate the Least Significant Difference (LSD) test.

Answer:- Rep

37. Write down the formula of combined or pooled proportion of two samples.

Answer:- (Page 290)

$$\overbrace{p_c} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

38. If approximate value of class interval is 2.96 and range = 14.8 then find the number of classes.

Answer:- (Page 29)

$$Class\ interval = \frac{range}{number\ of\ classes}$$

$$2.96 = \frac{14.8}{number\ of\ classes}$$

$$number\ of\ classes = \frac{14.8}{2.96} = 5$$

جھوٹ انسان اور ایمان دونوں کا دشمن ہے

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40. Find the coefficient of variation (C.V) for the following price of a commodity.
Price (X): 8, 13, 18, 23, 30

Answer:-

X	**X^2**
8	**64**
13	**169**
18	**324**
23	**529**
30	**900**
$\sum X = 92$	$\sum X^2 = 1986$

$$\overline{X} = \frac{\sum X}{n} = \frac{92}{5} = 18.4$$

$$S = \sqrt{\frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2}$$

$$S = \sqrt{\frac{1986}{5} - \left(\frac{92}{5}\right)^2}$$

$$S = \sqrt{397.2 - 338.56} = \sqrt{58.64} = 7.66$$

$$C.V = \frac{S}{\overline{X}} \times 100$$

$$C.V = \frac{7.66}{18.4} \times 100 = 41.63\%$$

41. Flaws in plywood occur at random with an average of one flaw per 50 square feet. What is the probability that 32 square feet will have no flaws?

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

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2- Probability of one electric device failure is 0.01. If a sample is chosen of 400, what is probability that exactly 2 are defective (Marks 5)

Answer:- (Page 218)

$p = 0.01$

$q = 1 - p \Rightarrow q = 1 - 0.01 \Rightarrow q = 0.99$

$$P(X = 2) = \binom{400}{2} (0.01)^2 (0.99)^{400-2}$$

$$= 79800 * 0.0001 * 0.02$$

$$= 0.1596$$

4- Two dices are rolled. Find probability that outcome is equal or more than 11. (Marks 5)

Answer:-

$$S = \begin{pmatrix} (1,1) & (2,1) & (3,1) & (4,1) & (5,1) & (6,1) \\ (1,2) & (2,2) & (3,2) & (4,2) & (5,2) & (6,2) \\ (1,3) & (2,3) & (3,3) & (4,3) & (5,3) & (6,3) \\ (1,4) & (2,4) & (3,4) & (4,4) & (5,4) & (6,4) \\ (1,5) & (2,5) & (3,5) & (4,5) & (5,5) & (6,5) \\ (1,6) & (2,6) & (3,6) & (4,6) & (5,6) & (6,6) \end{pmatrix}$$

$$P = (5,6), (6,5), (6,6)$$

$$= \frac{3}{36} = \frac{1}{12} = 0.83$$

5- If X is binomial distribution n=5, p=0.5, q=0.5 then find SD(X) (Marks 3)

Answer:- (Page 214)

$$S.D(X) = \sqrt{npq}$$

$$= \sqrt{5 \times 0.5 \times 0.5} = \sqrt{1.25}$$

$$= 1.12$$

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

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6- n=24, Mean=33, s=15, x=40.4. Compute t statistic (Marks 3)

7- 90% confidence interval of population mean is 11 to 20. Interpret result (Marks 3)
Answer:- Rep

8- E(XY)=421, E(X)=42, E(Y)=15. Check independence (Marks 3)
Answer:-

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$
$$= 421 - (42)(15)$$
$$= -209$$

It is not independent.

9- Difference between statistics and statistic (Marks 3)
Answer:-
Statistics is the plural of statistic, the science that deals with the collection, classification, analysis, and interpretation of numerical facts or data. While the statistic is a single value or piece of data.

10- Why we call standard deviation the standard error (Marks 2)
Answer:- (Page 240)
The square root of the variance is the standard deviation, and the standard deviation of a sampling distribution is termed as its standard error.

11- if E(Y)=0.5 then find E(10.5+2Y) ... (Marks 2)
Answer:-

$$= E(10.5 + 2Y)$$
$$= 10.5 + 2E(Y)$$
$$= 10.5 + 2 * 0.5$$
$$= 11.5$$

12- What is the meaning of 'b' in Y=a+bX (Marks 2)
Answer:- (Page 121)
b represents the slope of the line

13- Explain level of significance (Marks 2)
Answer:- rep

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

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Q# 28: can all deciles be expressed as percentile? Explain (2)
Answer: Yes all deciles cab be expressed as percentiles
As the D1 is 10 percent of data and P10 is 10% of data.
D2 is 20% of data whereas in percentile it is P20 and so on …

Q# 29: what is meant by sampling distribution? (2)
Answer: Page 237
The probability distribution of any statistic (such as the mean, the standard deviation, the proportion of successes in a sample, etc.) is known as its sampling distribution.

Q# 30: The department claims that the exceeds Rs. 2500 at the 0.05 level, then formulate null alternative hypothesis?(2)
Answer: Page 283
$H_0 : \mu < 2500$
$H_1 : \mu > 2500$

Q# 31: How we decide that the drawn sample is "Small Sample " or a "Large Sample"?(2)
Answer: rep

Q# 32: If E(X)=0.7 then find E(2X)? (2)
Answer:
$E(2X) = 2E(X)$
$= 2 * 0.7$
$= 1.4$

Q# 33: State the Baye's theorem? (3)
Answer: Page 166
If events A1, A2… Ak form a PARTITION of a sample space S (that is, the events Ai are mutually exclusive and exhaustive (i.e. their union is S)),
and if B is any other event of S such that it can occur ONLY IF ONE OF THE Ai OCCURS, then for any i,

$$P(A_i / B) = \frac{P(A_i)P(B/A_i)}{\sum_{i=1}^{k} P(A_i)P(B/A_i)}$$

Q# 35: "The 95% confidence interval for population mean is 1.3 to 4.7". Interrupt this result (3)
Answer: We are 95% sure that the population mean lies between 1.3 to 4.7

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

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Q# 37: If E(XY)=045 E(X)=0.50 and E(Y)=0.90 then X and Y are independent?(3)

Answer:

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$
$$= 0.45 - (0.50)(0.90)$$
$$= 0.45 - 0.45$$
$$= 0(\text{Therefore it is independent})$$

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2 Marks

2) If we draw a card from an ordinary deck of 52 cards. Can king and diamond be mutually exclusive events? Give reason to support your answer.

Answer:- (Page 146)

No, because If the two events *can* occur at the same time, they are not mutually exclusive, Therefore, kings and diamonds are not mutually exclusive.

3) Suppose that we toss a fair coin three times. What is its sample space?

Answer:- {(H,H,H),(H,T,H),(H,H,T),(T,H,H),(T,T,H),(T,H,T),(H,T,T),(T,T,T)}

4) Why we take B_2 =3 as the criteria for measuring the kurtosis of any distribution?

Answer:- (Page 228)

For the normal distribution, β2 comes out to be 3, this is why this value has been taken as a criterion for measuring the kurtosis of any distribution.

5) Define level of significance?

Answer:- rep

6) How we decide that the drawn sample is "Small sample" or a "large sample"?

Answer:- Rep

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

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3 Marks

7) If mean of a distribution is 0.925 and standard deviation is 0.132. Find out the limits by applying chebychev's inequality for k=2. How much fraction of data will lie between the two limits?

Answer:- (Page 95)

$$\left(\overline{X}-kS,\overline{X}+kS\right)$$

$$\left(0.925-2\left(0.132\right),0.925+2\left(0.132\right)\right)$$

$$\left(0.925-0.264,0.925+0.264\right)$$

$$\left(0.661,1.189\right)$$

95% of the measurements will fall within these limits.

8) If E(XY)=7.5, E(X)=2.4 and E(Y)=4.3, calculate covariance of X and Y.

Answer:-

$$\begin{aligned} Cov\left(X,Y\right) &= E\left(XY\right)-E\left(X\right)E\left(Y\right) \\ &= 7.5-\left(2.4\right)\left(4.3\right) \\ &= 7.5-10.32 \\ &= -2.82 \end{aligned}$$

11) Write down the two properties of sampling distribution of proportion p', when the sampling is performed with replacement?

Answer:- (Page 241)

Property 1: $\mu_{\overline{X}}=\mu$

Property 2: In case of sampling with replacement:

$$\sigma_X=\frac{\sigma}{n}$$

12) What is natural pairing in observations? Give example.

Answer:- (Page 302)

Natural pairing occurs whenever measurement is taken on the same unit or individual at two different times. For example, suppose ten young recruits are given a strenuous physical training programme by the Army. Their weights are recorded before they begin and after they complete the training. The two observations obtained for each recruit i.e. the before-and-after measurement constitute natural pairing. The above is natural pairing.

بد صورت چہرہ بد صورت دماغ سے بہتر ہے

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5 Marks

13) Interpret the concept of five numbers summary and also explain the purpose of five number summaries.

Answer:- (Page 97)

A five-number summary consists of X0,Q1, Median, Q3, and Xm ; It provides us quite a good idea about the shape of
the distribution
If the data were perfectly symmetrical, the following would be true:
The distance from Q1 to the median would be equal to the distance from the median to Q3
The distance from X0 to Q1 would be equal to the distance from Q3 to Xm.
The median, the mid-quartile range, and the midrange would all be equal. All these measures would also be equal to the arithmetic mean of the data

On the other hand, for non-symmetrical distributions, the following would be true:
In right-skewed distributions the distance from Q3 to Xm greatly exceeds the distance from X0 to Q1
in right-skewed distributions,
median < mid-quartile range < midrange:
THE POSITIVELY SKEWED CURVE

14) What is the probability that a poker hand of 5 cards contain exactly 2 aces (hyper geometric distribution)?

Answer:-

$$P(X=2)=\frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}}$$

$$=\frac{6\times 17296}{2598960}=\frac{103776}{2598960}=0.0399$$

16)From the table given below find the value of chi square

Obvervation Frequency Oij	Expected Frequency Eij
120	100
130	150
80	100
170	150

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the mean of a population is 6 a sample of size '20' is taken with replacement from this population.which one of the following is the mean of sampling distribution of mean.

1) 2 2) 8 3) 6 4) 3

Answer:-

6

Q no 2 Write down the significance of experimental design? marks :2

Answer:- Rep

Q No: 03 Explain the formula of hyper geometric probability distribution. marks :2

Answer:- (Page 219)

$$P(X=x)=\frac{\binom{k}{n}\binom{N-k}{n-x}}{\binom{N}{n}}$$

Where

N = number of units in the population,

n = number of units in the sample, and

k = number of successes in the population.

The hyper geometric probability distribution has three parameters N, n and k.

Q no 5

Suppose that we toss a fair coin three times. What is the sample space? marks :2

Answer:- Rep

Compute the following formula which is used to find the sample size.

Answer:- For finding sample space of coin the formula is: $2^n = 2^3 = 8$

If x is a passion random variable with parameter value '4.5', find p(x=1)? Marks: 03

Answer:- (Page 222)

$$P(X=1)=\frac{e^{-4.5}(4.5)^1}{1!}$$
$$=0.0499$$

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Q no: 09
When the value of chi-square is equal to zero? Marks : 03
Answer:- (Page 338)
If the observed frequencies are exactly equal to the expected ones, then chi-square will be exactly equal to zero.

Q no: 10 If x= 255, n=634, Po=0.60 then find the z-test statistics for probability. Marks: 03
Answer:- (Page 289)

$$Z = \frac{X \pm \frac{1}{2} - np_0}{\sqrt{np_0(1-p_0)}}$$

$$np_0 = 634 * 0.60 = 380.4$$

$$here\ X < np_0\ therefore\ we\ use\ X + \frac{1}{2}$$

$$Z = \frac{X + \frac{1}{2} - np_0}{\sqrt{np_0(1-p_0)}}$$

$$Z = \frac{225 + \frac{1}{2} - 380.4}{\sqrt{380.4(1-0.60)}}$$

$$Z = \frac{-154.9}{12.34}$$

$$Z = -12.55$$

STA301- Statistics and Probability
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Describe the formula for hypergeometric distribution? 2 marks
Answer:- Rep
What is graphical representation? 2 marks
Answer:-
The plot of the points in the plane which constitute the graph of a given real function or a pictorial diagram depicting interdependence of variables.

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What is the t formula for paired distribution? 3marks.

Answer:- (Page 302)

$$t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}}$$

What is the probability that at least one head occurs if a coin tossed 6 times successively? 5 marks

Answer:-

$$P(X = 1) = \binom{6}{1} \cdot \left(\frac{1}{2}\right)^1 \cdot \left(\frac{1}{2}\right)^{6-1}$$

$$= \binom{6}{1} \cdot \left(\frac{1}{2}\right)^1 \cdot \left(\frac{1}{2}\right)^5$$

$$= 3.03$$

Find first 2 moments for function with respect to mean? 5 marks

Answer:- (Page 192)

16)From the table given below find the value of chi square

Obervation Frequency Oij	**Expected Frequency Eij**
120	**100**
130	**150**
80	**100**
170	150

Answer:- Rep

STA301- Statistics and Probability

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Q no 5

Suppose that we toss a fair coin three times. what is the sample space? marks :2

Answer: Rep

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو

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Compute the following formula which is used to find the sample size.

$$n = \left(\frac{Z_{\alpha/2}}{e} \right)$$

Page 276

FINALTERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 4)

Question No: 31 (Marks: 2)
What is the mean and variance of Poisson distribution?
Answer:- (Page 233)
If the random variable X has a Poisson distribution with parameter μ, then its mean and variance are given by E(X) = μ and Var(X) = μ.

Question No: 32 (Marks: 2)
Explain the Chi-square test of goodness of fit.
Answer:- (Page 332)
The chi-square test of goodness-of-fit is a test of hypothesis concerned with the comparison of observed frequencies of a sample, and the corresponding expected frequencies based on a theoretical distribution.

Question No: 33 (Marks: 2)
Define level of significance?
Answer:- Rep

Question No: 34 (Marks: 3)
Find the value of F(table value), when $n_1 = 7$, $n_2 = 10$ and $\alpha = 0.05$
Answer:- (Page 316)

Question No: 35 (Marks: 3)
Define Null and Alternative hypothesis.
Answer:- Rep

خوبصورتی علم وادب سے ہوتی ہے لباس و حسن سے نہیں

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Question No: 36 (Marks: 3)
A random sample of 100 is taken from a population with mean 30 and standard deviation 5. The probability distribution of the parent population is unknown, find the mean and standard error of the sampling distribution of $\bar{X}$.

Question No: 37 (Marks: 5)
What is the probability that a poker hand of 5 cards contain exactly 2 aces (hypergeometric distribution)?
Answer:- Rep

Question No: 38 (Marks: 5)
A random sample of size n is drawn from normal population with mean 5 and variance σ^2.
If n=25, s=10 and t=2, what is the values of $\bar{x}$?

Question No: 39 (Marks: 5)
Describe the main steps of General Procedure for Testing Hypothesis.
Answer:- (Page 281)
Step-1:
Formulation of the Null and Alternative Hypotheses:
Step-2:
Decision Regarding the Level of Significance
Step-3:
Test Statistic (that statistic that will enable us to test our hypothesis):
Step-4:
Calculations:
Step-5:
Critical Region (that portion of the X-axis which compels us to reject the null hypothesis):
Step-6:
Conclusion:

FINALTERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 4)

Question No: 31 (Marks: 2)
How many parameters are involved in hyper geometric distribution?
Answer:- (Page 291)
The hyper geometric probability distribution has three parameters N, n and k.

تم اچھا کرو زمانہ تم کو برا سمجھے یہ اس سے بہتر ہے کہ تم برا کرو اور زمانہ تم کو اچھا سمجھے

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Question No: 32 (Marks: 2)
If an automobile is driven on the average no more than 16000 Km per year, then formulate the null and alternative hypothesis.
Answer:- Rep

Question No: 33 (Marks: 2)
Write down the test statistic when chi- square goodness of fit test is performed.
Answer:- (Page 334)

$$\chi^2 = \sum_i \frac{(o_i - e_i)^2}{e_i}$$

Question No: 34 (Marks: 3)

Find the value of F(table value), when $n_1 = 7$, $n_2 = 10$ and $\alpha = 0.05$
Answer:- (Page 316)
F0.05(n1-1, n2-1) = (6,9) =3.37
F0.05(n2-1, n1-1) = (9,6)
V2=6 , v1=8 => 4.15,
V2=6, v1=12 => 4.00
Mean = (4.15+4)/2 = 4.075
Hence value of F0.05(9,6) = 4.075

Question No: 35 (Marks: 3)

If X = 327, n = 634, $p_0 =$ 0.50 then find the z-test statistic for proportion.
Answer:- (Page 289)

$$Z = \frac{X \pm \frac{1}{2} - np_o}{\sqrt{np_o(1-p_o)}}$$

$$np_o = 634 * 0.50 = 317$$

$$here\ X > np_o\ therefore\ we\ use\ X - \frac{1}{2}$$

$$Z = \frac{X - \frac{1}{2} - np_o}{\sqrt{np_o(1-p_o)}}$$

$$Z = \frac{327 - \frac{1}{2} - 317}{\sqrt{317(1-0.50)}}$$

$$Z = \frac{9.5}{12.58}$$

$$Z = 0.75$$

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Question No: 36 (Marks: 3)
If population proportions are given as:

$$P_1 = 0.30,\ P_2 = 0.20.$$

Find $\sigma^2_{\hat{p}_1 - \hat{p}_2}$,where n = 10

Answer:- (Page 256)

$$\sigma_{\hat{p}1-\hat{p}2} = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}$$

$$q_1 = 1 - p_1 \Rightarrow 1 - 0.30 = 0.7$$

$$q_2 = 1 - p_2 \Rightarrow 1 - 0.20 = 0.8$$

$$\sigma_{\hat{p}1-\hat{p}2} = \frac{(0.30)(0.7)}{10} + \frac{(0.20)(0.80)}{10}$$

$$= 0.021 + 0.016$$

$$= 0.037$$

Question No: 37 (Marks: 5)
A candidate for mayor in a large city believes that he appeals to at least 10 per cent more of the educated voters than the uneducated voters. He hires the services of a poll-taking organization, and they find that 62 of 100 educated voters interviewed support the candidate, and 69 of 150 uneducated voters support him at the 0.05 significance level.
Answer:- (Page 292)
Step-1:
The null and alternative hypothesis is
H0 : p1 – p2 > 0.10, and
H1 : p1 – p2 < 0.10, where p1 = proportion of educated voters, and p2 = proportion of uneducated voters.

Step-2:
Level of Significance:
α = 0.05.

انسان کے لئے بری صحبت سے بڑھ کر بری کوئی چیز نہیں

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Step-3:

Test Statistic:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0.10}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}$$

which for large sample sizes, is approximately standard normal.

Step-4:

Computation:

$$\text{Here} \quad \hat{p}_1 = \frac{62}{100} = 0.62, \text{ so that } \hat{q}_1 = 0.38,$$

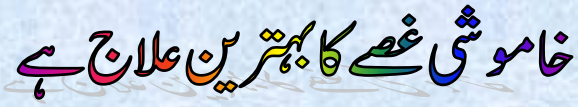
$$\hat{p}_2 = \frac{69}{150} = 0.46, \text{ so that } \hat{q}_2 = 0.54.$$

$$\text{Thus} \quad z = \frac{(0.62 - 0.46) - 0.10}{\sqrt{\frac{(0.62)(0.38)}{100} + \frac{(0.46)(0.54)}{150}}}$$

$$= \frac{0.06}{\sqrt{0.002356 + 0.001656}} = \frac{0.06}{0.063} = 0.95.$$

Step-5:
Critical Region:
As this is a one-tailed test, therefore the critical region is given by
Z < -z0.05 = -1.645

Step-6:
Conclusion:
Since the calculated value z = 0.95 does not fall in the critical region, so we accept the null hypothesis H0 : p1 – p2 > 0.10.The data seems to support the candidate's view.
Until now, we have discussed in considerable detail interval estimation and hypothesis-testing based on the standard normal distribution and the Z-statistic.
Next, we begin the discussion of interval estimation hypothesis-testing based on the t-distribution.



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Question No: 38 (Marks: 5)
If we have RCBD with MSE=3.19, no.of.treatments = 4, no.of.blocks = 5; then find the value of LSD (least significant difference) for treatments by using α=0.05 and error degrees of freedom is 12.
Answer:- Rep

Question No: 39 (Marks: 5)
Find the mean and variance for the sampling distribution given below.

$(\hat{p})$	No. of Samples	Probability $f(\hat{p})$
0	1	1/20
1/3	9	9/20
2/3	9	9/20
1	1	1/20
Σ	20	1

Answer:- Rep

FINALTERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 3)

Question No: 31 (Marks: 2)
Define normal distribution.
Answer:- (Page 226)
A continuous random variable is said to be normally distributed with mean μ and standard deviation σ if its probability
density function is given by

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}$$

Question No: 32 (Marks: 2)
Fill up the missing values in the formula.

$$n=\frac{(\quad_{\alpha/2})^2\ \hat{p}\hat{q}}{\quad}$$

جھوٹ رزق کو کھا جاتا ہے

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Answer:- (Page 277)

$$n = \frac{\left(z_{\alpha/2}\right)^2 \hat{p}\,\hat{q}}{e^2}$$

Question No: 33 (Marks: 2)
Explain the Chi-square test of goodness of fit.
Answer:- Rep

Question No: 34 (Marks: 3)
What is natural pairing in observations? Also give an example.
Answer:- Rep

Question No: 35 (Marks: 3)
If population proportions are given as:

$$P_1 = 0.30,\ P_2 = 0.20.$$

Find $\sigma^2_{\hat{p}_1 - \hat{p}_2}$,where n = 10

Answer:- Rep

Question No: 36 (Marks: 3)
Explain the method of Maximum Likelihood in Point Estimation?
Answer:- Page 263
The method of maximum likelihood is regarded as the *MOST important* method of estimation, and is the *most* widely used method. This method was introduced in 1922 by Sir Ronald A. Fisher (1890-1962).The mathematical technique of finding Maximum Likelihood Estimators is a bit *advanced*, and involves the concept of the Likelihood Function.

Question No: 37 (Marks: 5)
The following data was obtained for a randomized block design involving five treatments and three blocks SST=430, SSTR=310, SSB=85, Setup the ANOVA table.
Answer:- (Page 332)

Source	d.f	SS	MS
Between treatments	4	430	107.5
Between Blocks	2	85	42.5
Error	8	310	38.75
Total	14	825	188.75

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Question No: 38 (Marks: 5)
In a Standard Normal Distribution, find:
Mean and Standard Deviation
Lower Quartile
Upper quartile
Inter-quartile range
Mean Deviation

Answer:-
In normal distribution

i. Mean and Standard Deviation
Mean is 0 and S.D is 1

ii. Lower Quartile:
$Q_1 = \mu - 0.6745\sigma$
=0-0.6745(1)
Q_1 **= -0.6745**

iii. Upper Quartile:
$Q_3 = \mu + 0.6745\sigma$
=0.6745

iv. Inter-quartile range
$Q_3 - Q_1/2$ **= 0.6745 - (-0.6745)/2 = 0.6745**

v. Mean Deviation
= $\mathbf{0.7979}\sigma$
= 0.7979(1)
= 0.7979

Question No: 39 (Marks: 5)
If $n = 13, \bar{x} = 34, \sigma^2 = 70, \alpha = 0.10$
Test the hypothesis $\mu > 31$

افضل انسان وہ ہے جو اپنی اصلاح کی کوشش کرتا ہے

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Answer:-

$H_0 = \mu = 30$

$H_1 = \mu > 30$

Level of significance α =0.10

α /2=0.05

Critical region=z>1.645

$$z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$Z = \frac{34 - 31}{2.32}$$

Z=1.29

Conclusion: Since the value Z=1.29 is less than the value of critical region so we accepted H_0 at 0.10 Level of significance.

FINALTERM EXAMINATION
Fall 2009
STA301- Statistics and Probability (Session - 4)

Question No: 21 (Marks: 1)
Write down the formula for binomial distribution.

Answer:- (Page 212)

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

Question No: 22 (Marks: 2)
Write down the formula for testing the equality of two population proportions.

Answer:- (Page 290)

$$Z = \frac{\left(\hat{p}_1 - \hat{p}_2\right) - 0}{\sqrt{\hat{p}_c \hat{q}_c \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

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Question No: 23 (Marks: 3)
Define moment ratios. In which unit they are expressed?

Answer:- (Page 119)

$$b_1 = \frac{(m_3)^2}{(m_2)^3} \quad and \quad b_2 = \frac{m_4}{(m_2)^2}$$

They are independent of origin and units of measurement, i.e. they are pure numbers.

ہر چیز کی ایک پہچان ہوتی ہے اور عقلمند کی پہچان غور و فکر کرنا ہے اور غور و فکر کی پہچان خاموشی ہے

اس سے پہلے کہ تمہیں شہوت فتنے میں ڈالے نکاح کر لو

اپنی مرضی اور اللہ کی مرضی میں فرق کا نام غم ہے

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